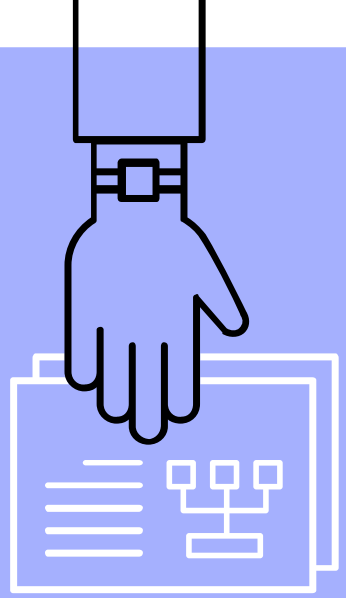
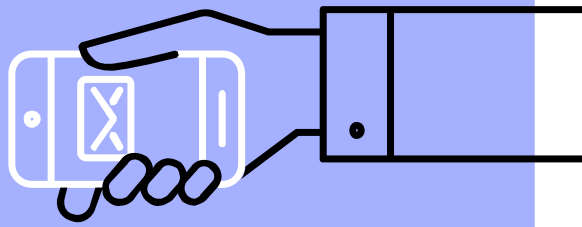
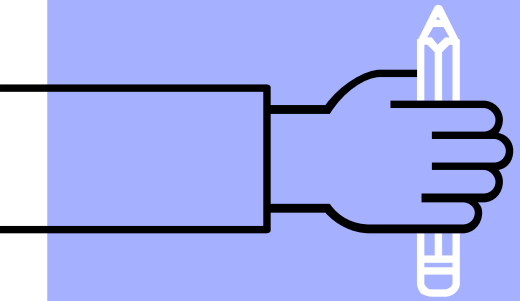
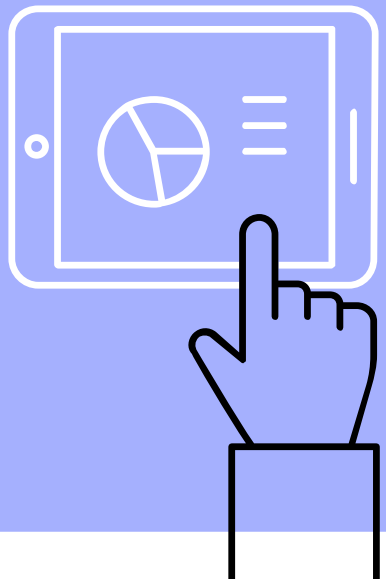
Growth Modeling & Data-Driven Strategy







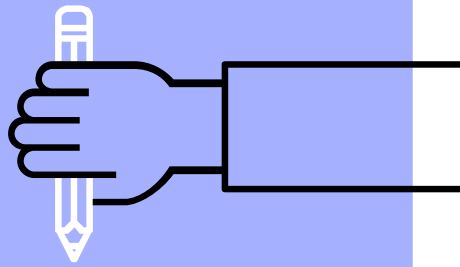


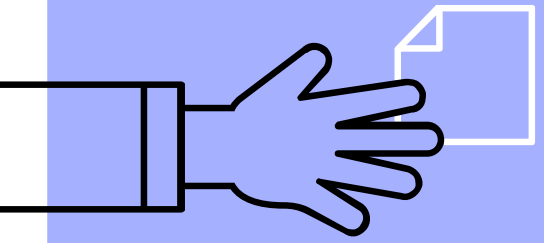
HELLO!

I am Aurore Kervella

I have been working in Data and Growth for more than 15 years at companies like Dior, Sephora, Uber, Heetch.

1.
THE ARCHITECTURE OF GROWTH





Funnels describe
Loops drive

Funnels

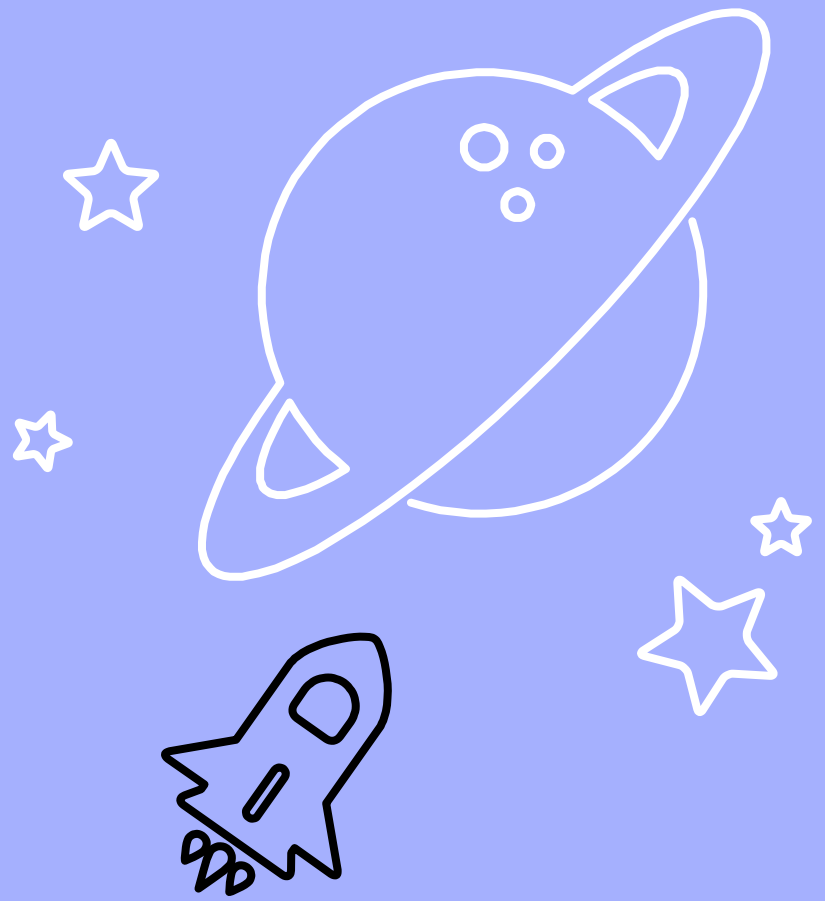
the anatomy of conversion

Loops

the mechanics of compounding

North Star Metric

the guiding light



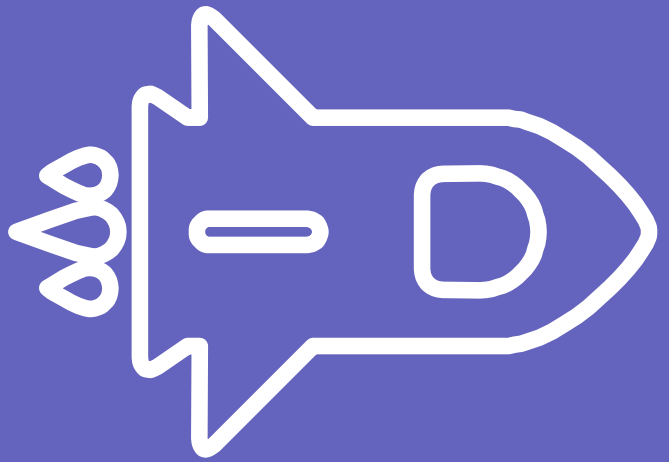
WHAT IS GROWTH REALLY?

Marketing

Loops

Data

Product





Growth

=

increase in users, revenue, or engagement over time



WHAT IS GROWTH?

- **Linear** = fixed increments
  Steady, predictable increases in fixed increments. Think traditional advertising - spend more, get proportionally more results.

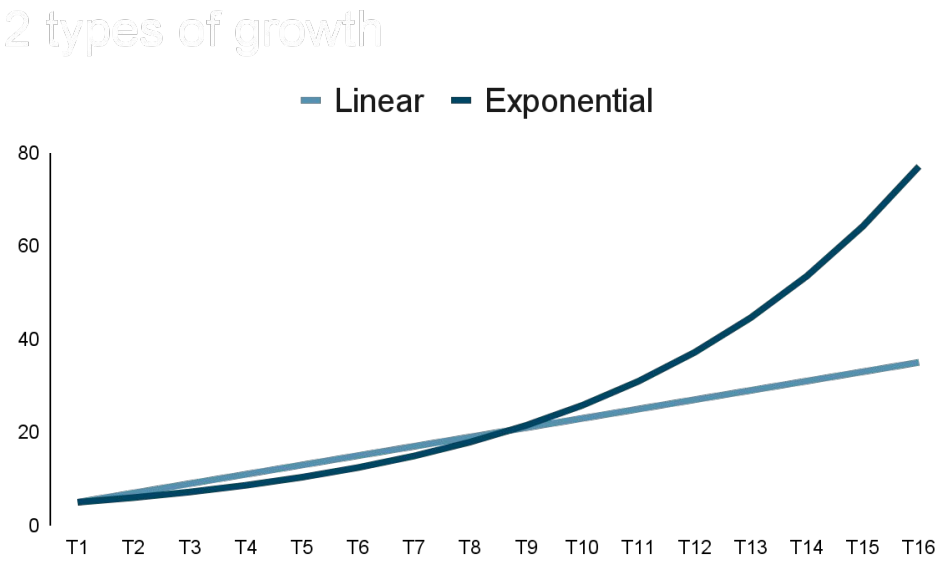
  $y = a + b * x$

- **Exponential** = compounding
  Compounding effects where growth accelerates over time. Each new user potentially brings more users.

  $y = a * b \wedge x$



2 TYPES OF GROWTH



"Why do most startups seek exponential rather than linear growth? What makes it harder to achieve?"

THE ENGINE OF GROWTH

- **Funnels**
  A multi-step process (e.g. AARRR) where users drop off at each step. Requires **continuous input** (marketing spend) at the top.

- **Loops**
  A closed system where the **output** of one cycle (e.g., a retained user) directly feeds the **input** of the next cycle (e.g., new user acquisition). Drives compounding growth.



UNDERSTANDING FUNNELS

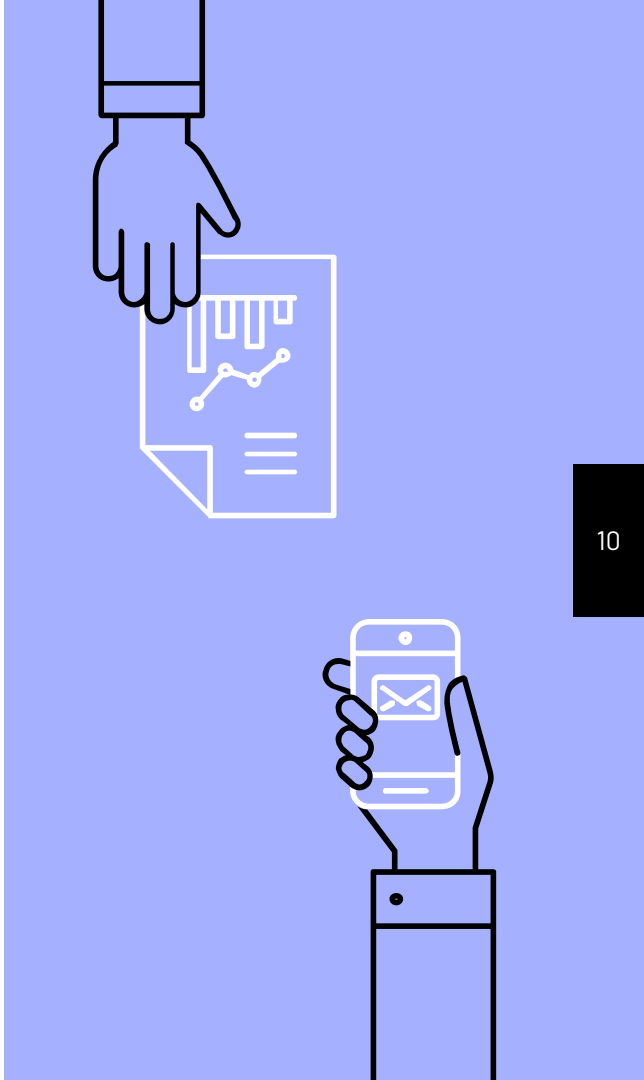
The traditional linear approach to converting prospects into customers through sequential stages.

Awareness
Potential customers discover your product

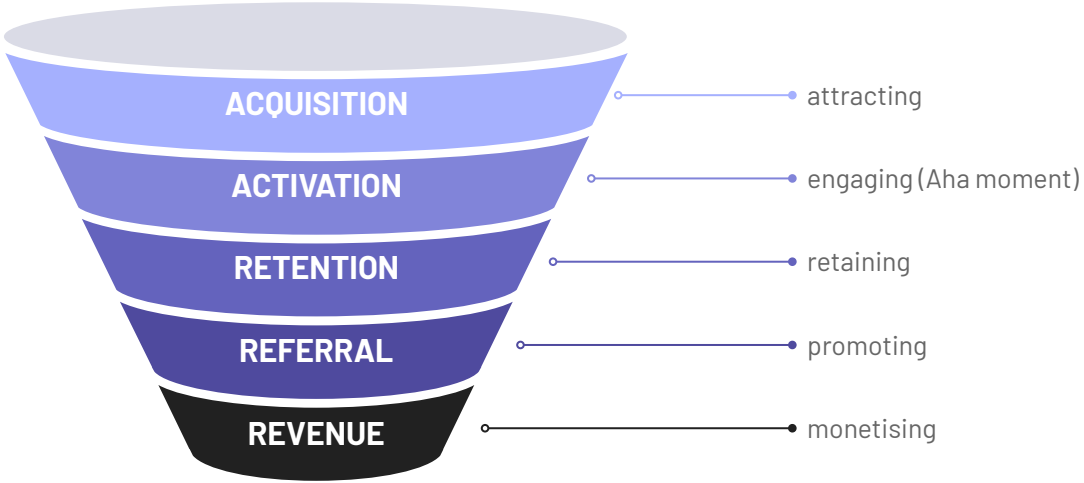
Consideration
Users evaluate your solution against alternatives

Conversion
Prospects become paying customers

Key Focus: Optimizing efficiency and identifying where users drop off in the process.

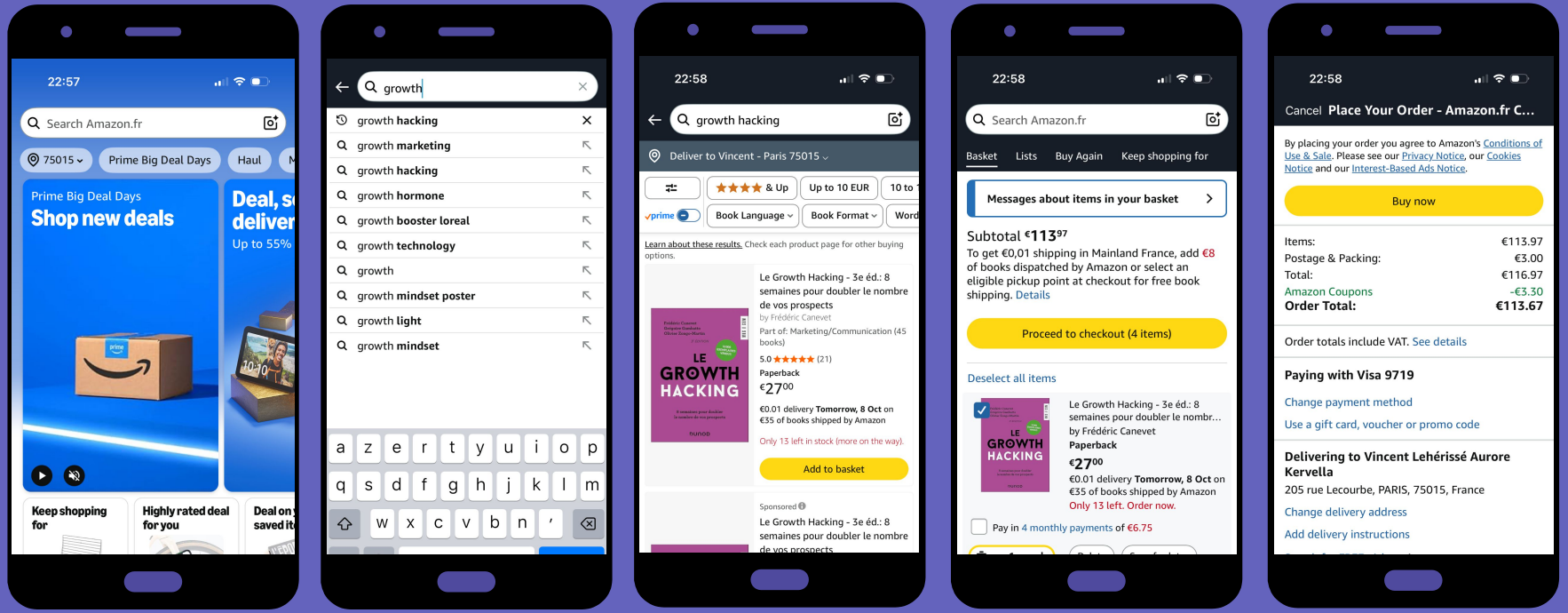


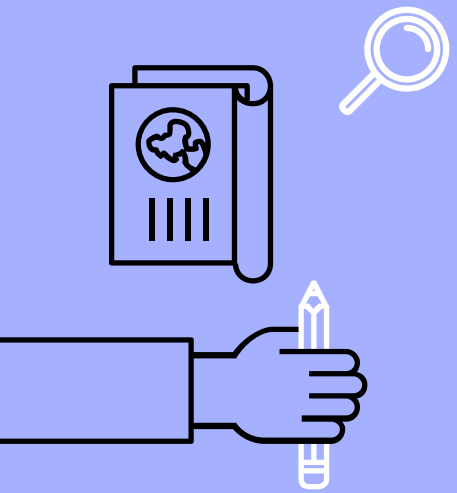
THE CLASSIC FUNNEL : AARRR



Conversion × Retention × Reactivation = Growth

Funnel example : Amazon





Funnel exercise

- You’re launching a new product : a learning app (Duolingo like)
- Your objective is to **convert users into paying subscribers**
- Define the **4 key steps** of the growth funnel from acquisition to retention / revenue



Funnels measure efficiency

- Linear
- Descriptive
- Optimizes conversion
- Ends with sale



Loops create momentum

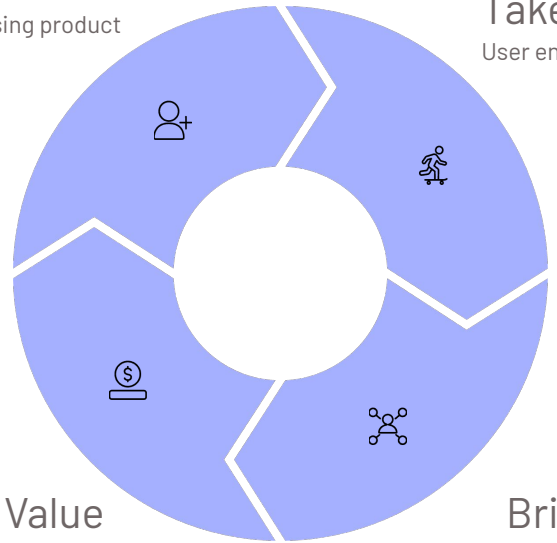
- Circular
- Generative
- Sustains momentum
- Feeds itself

GROWTH LOOPS EXPLAINED

Self-reinforcing systems where existing users drive acquisition of new users, creating compounding growth.

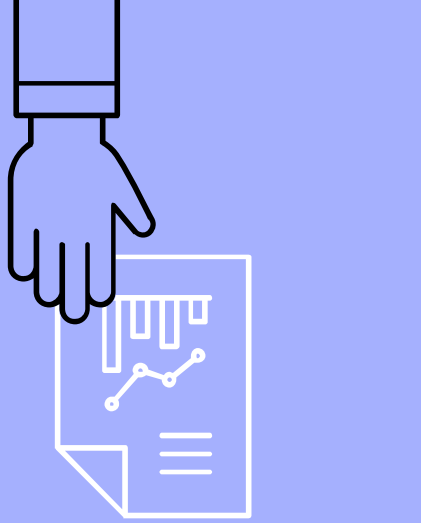
User Joins
New customer starts using product

Takes Action
User engages with core features



Creates Value
More users = more value for everyone

Brings New Users
Action naturally involves others





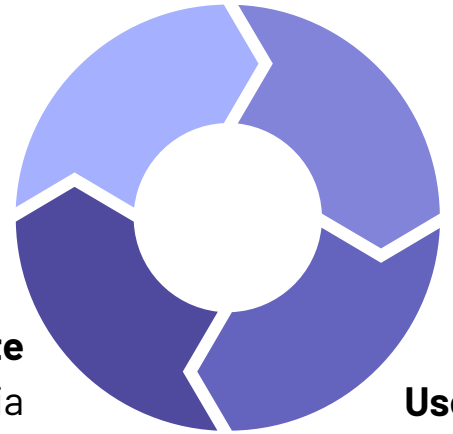
Type	Core Mechanism	Example	Key Metric(s)	Underlying Psychology / Engine
Acquisition Loop	Every user helps bring in *new users* — direct virality or content exposure.	**Dropbox** → Users invite friends to earn free storage. **TikTok** → Every piece of content shared externally drives new installs.	**K-Factor**, Invite Rate, Conversion Rate	Social proof, reciprocity, status — people share because it gives them utility or identity.
Retention Loop	Users stay engaged because *each action increases their own future value* — the more they use, the harder it is to leave.	**Duolingo** → Streaks, leaderboards, progress tracking. **Slack** → The more messages your team sends, the more integrated it becomes.	**DAU/MAU**, Churn Rate, Cohort Retention	Habit formation, sunk cost effect, community belonging.
Monetization Loop	User activity or purchases make the product *more valuable for others or more likely to generate revenue again.*	**Amazon** → Reviews → Trust → More Sales → More Reviews. **Freemium SaaS** → Free users convert → Revenue funds better product → More users convert.	**ARPU**, Conversion Rate, Net Revenue Retention (NRR)	Trust, network effect, economic flywheel.

Loop example : Dropbox

Dropbox referral program (extra storage for both inviter and invitee)

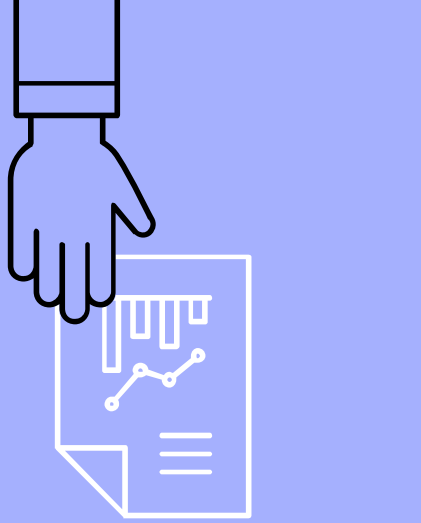
User signs up
start storing and sharing their files

Incentive to share
Extra storage if they refer their friends to join

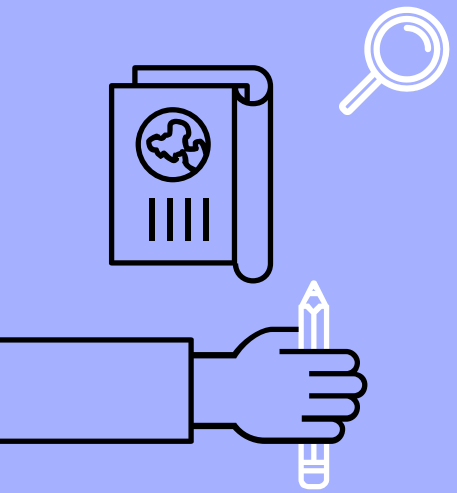


New user invite
referral invites sent via email or social media (nice and easy)

User select contacts
Invites are sent

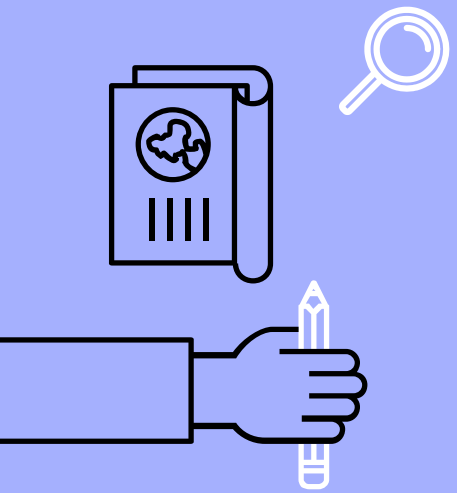






Loop exercise

- Design the primary growth loop for a platform
- We will use **LinkedIn** as our case study
  - Identify the Value
  - Identify the Input / Output
  - Map the Steps on the loop



Loop vs Funnel exercise

- Identify one funnel-driven company and one loop-driven company
- Share with the group



North Star Metric

(NSM)

=

A single metric that best captures the core value delivered to users.



THE NORTH STAR METRIC

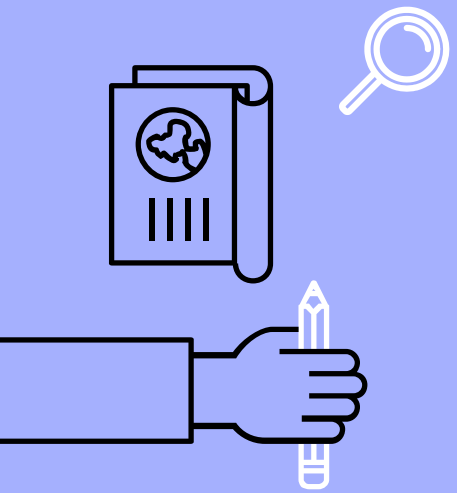
- **Aligns** cross-functional teams.
- Focuses on **sustainable, long-term growth** instead of vanity metrics.
- Typically a retention or engagement metric of some kind.
  - Daily active users (collaboration apps)
  - Monthly active subscribers (subscription)
  - Marketplace businesses : chicken or egg problem : usually supply side



THE NORTH STAR METRIC

- Examples :
  - Airbnb → Nights booked.
  - Spotify → Time spent listening.
  - Slack → Messages sent per team.



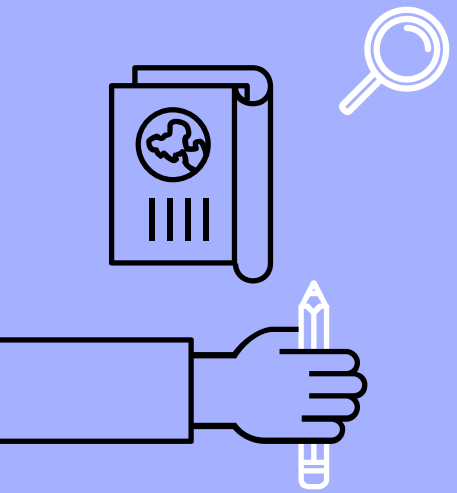


NSM exercise

- Revisit your team’s funnel or loop.
- What metric best reflects real value creation?
- Why would aligning everyone around it help?
- Share with the group

MINI RECAP BOARD

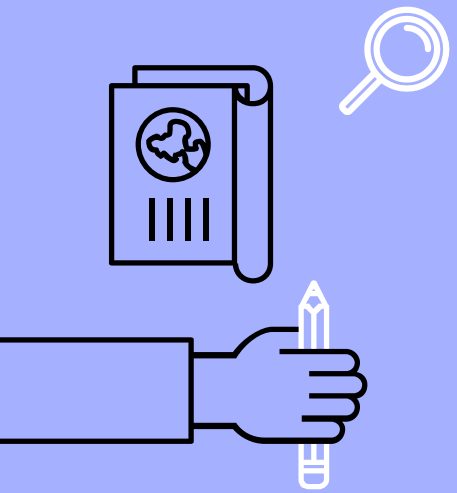
Concept	Purpose	Risk
Funnel	Diagnose journey	Over optimize micro steps
Loop	Driver compounding	Hard to identify early
NSM	Align teams	Tunnel vision



Time for a quizz (1/3)

You’re analyzing a startup that spends heavily on ads to drive installs.
Once users sign up, very few invite others, and growth only happens when ad spend increases.
Which growth architecture does this company use?

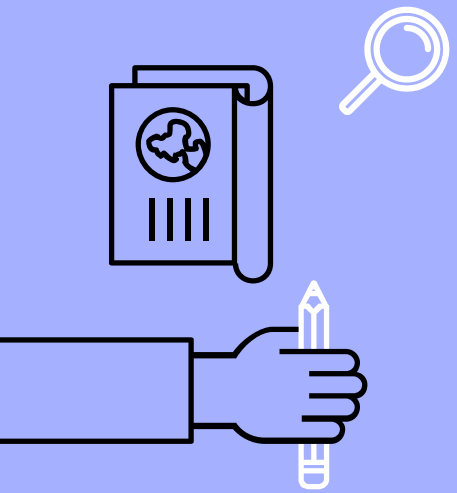
A. Loop
B. Funnel
C. Viral engine
D. Product-led system



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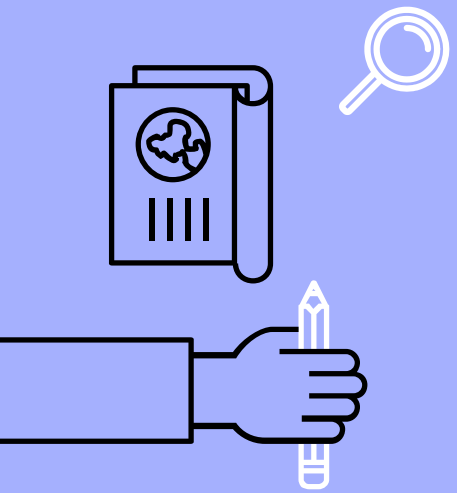
A. Loop
B. Funnel
C. Viral engine
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Time for a quizz (2/3)

Which of the following is the *best* North Star Metric for a food-delivery platform?

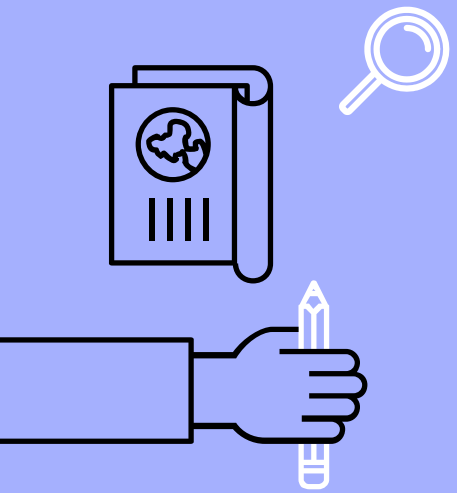
A. Number of app downloads
B. Total marketing spend
C. Completed deliveries per active customer
D. Number of push notifications sent



Time for a quizz (2/3)

Which of the following is the *best* North Star Metric for a food-delivery platform?

A. Number of app downloads
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Time for a quizz (3/3)

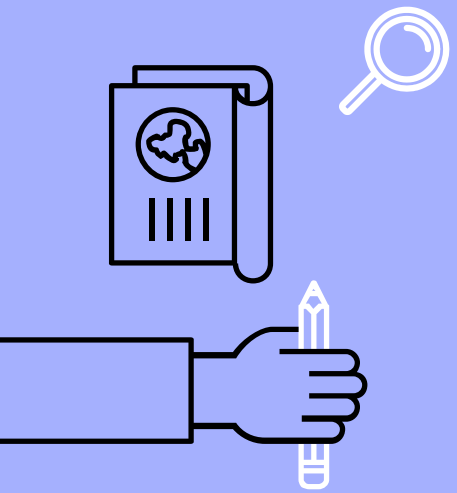
Dropbox famously grew through a referral program where users got extra storage for inviting friends. Which element made this a true growth loop rather than just a marketing tactic?

A. Each new user could trigger the same invite behavior.

B. Invited friends were automatically subscribed.

C. The reward was financial.

D. Dropbox ran TV ads about it.



Time for a quizz (3/3)

Dropbox famously grew through a referral program where users got extra storage for inviting friends. Which element made this a true growth loop rather than just a marketing tactic?

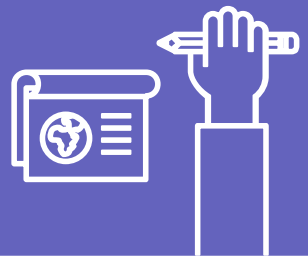
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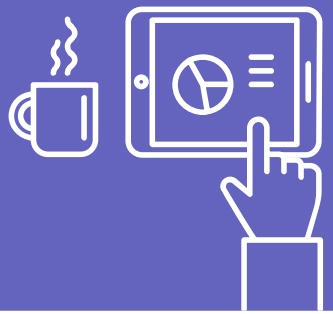
Which part of a growth system is hardest to measure?

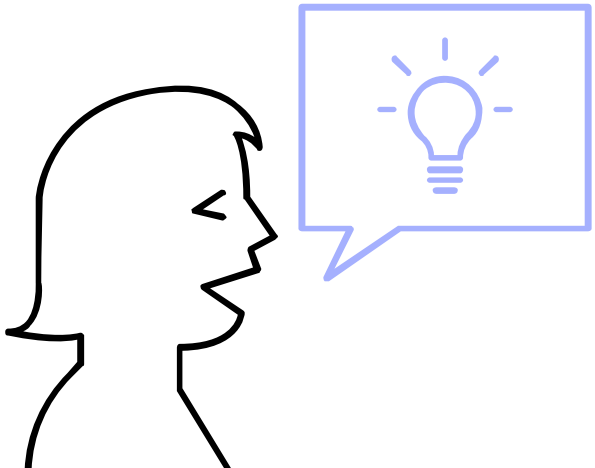




Can a company survive without loops?

When does a North Star conflict with user trust?





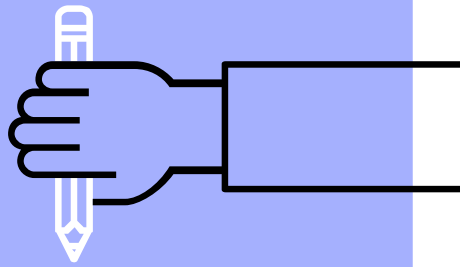
Growth can be linear or exponential

Loops drive compounding

Funnels = efficiency

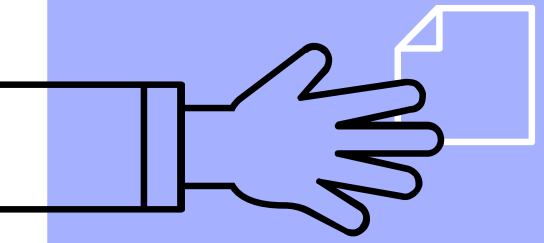
Loops = sustainability

NSM aligns teams and ensures value creation



2.
METRICS THAT MATTER

From data to decision



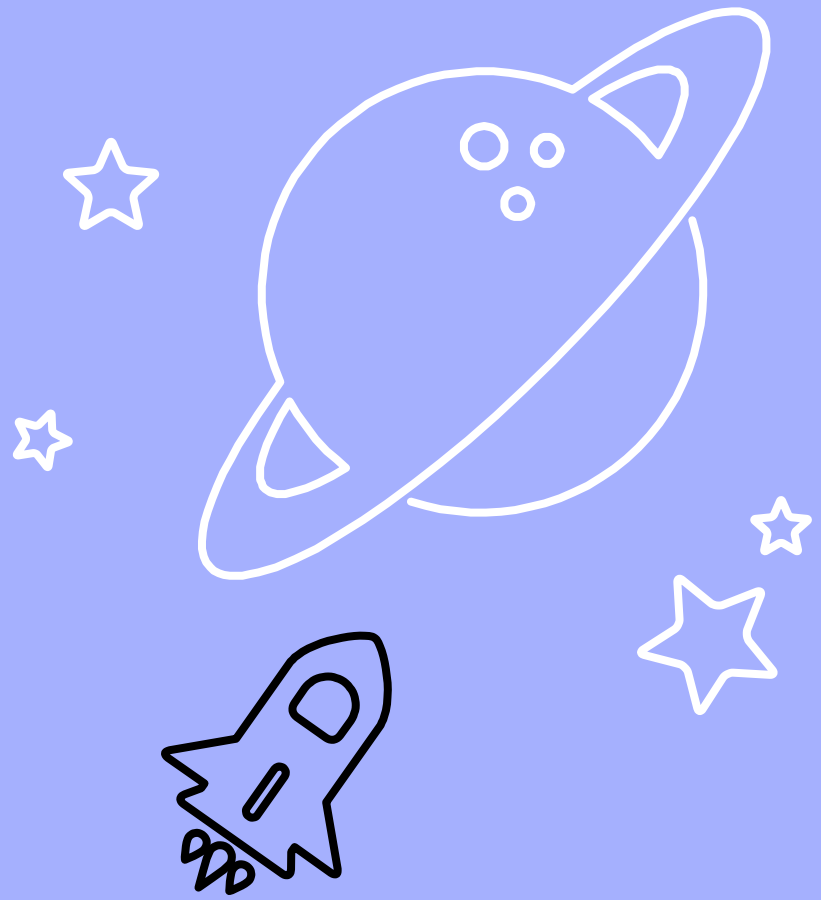
CAC & LTV

cost and revenue

Churn & Retention

How to retain a user

Virality & Word of Mouth



WHO IS REALLY GROWING?

Startup A



grows fast with paid ads (€1 M burn)

Startup B



grows slowly but retains 80 % monthly



Grow Metrics

(CAC, LTV,...)

=

Without the right metrics, growth looks like magic. With them, it becomes a science

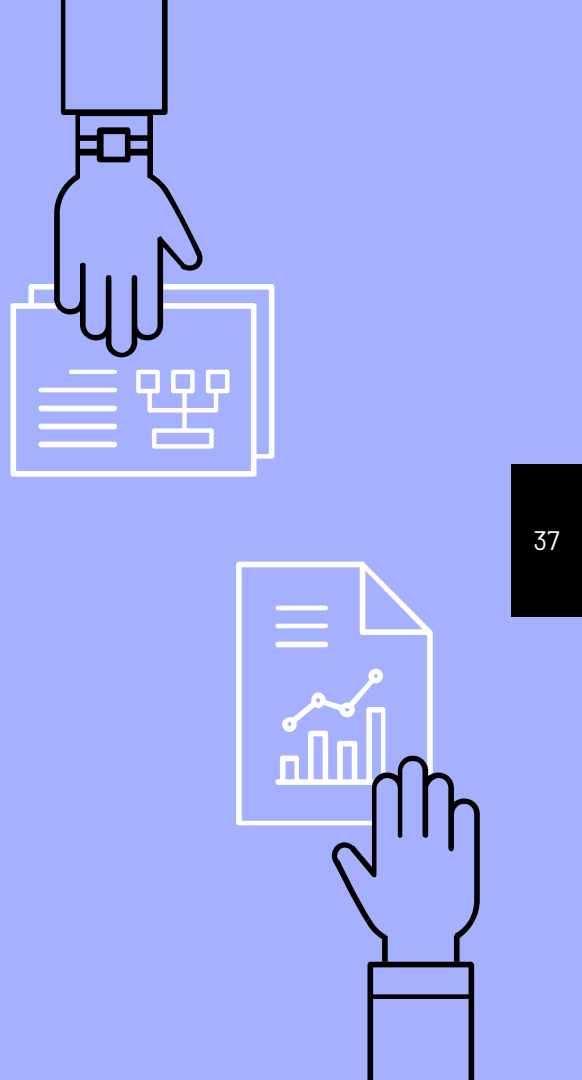


CAC (cost of acquisition)

How much it costs on average to acquire new clients ?

$$CAC = \frac{Total\,Acquisition\,Spend}{New\,Customers}$$

- Example :
  - Total spend: $10,000
  - New customers: 500
  - **CAC** = $10,000 / 500 = **$20 per customer**

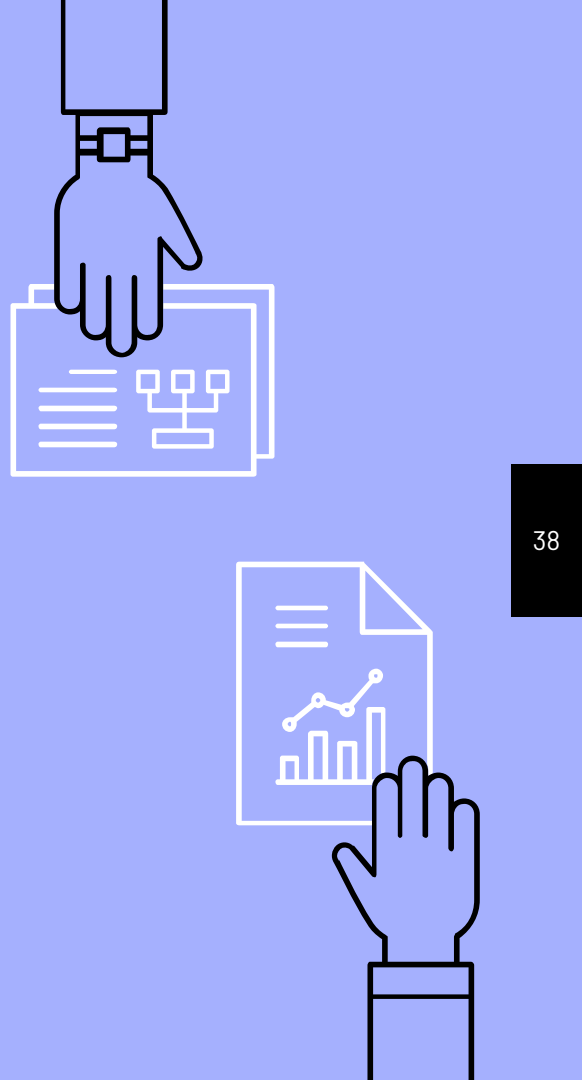


LTV (lifetime value)

The total net revenue you expect to earn from a customer before they churn.

$$LTV = \text{ARPU} \times \text{Gross Margin} \times \text{Customer Lifetime}$$

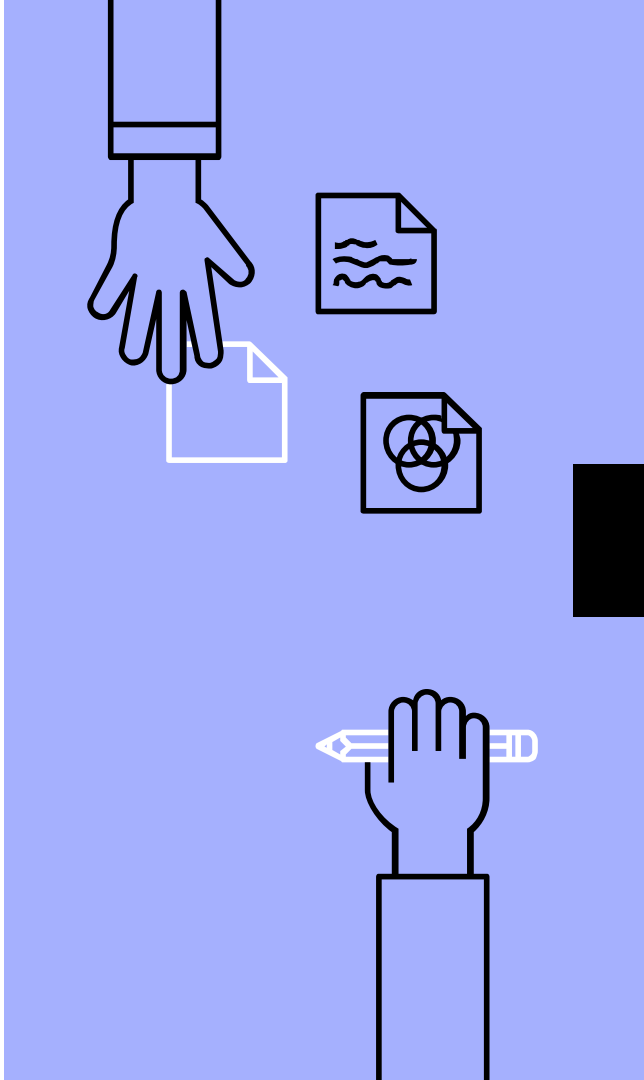
- Definitions :
  - **ARPU** = Average Revenue per User
  - **Gross Margin** = % of revenue retained after costs
  - **Customer Lifetime** = Average time before churn (can be replaced by 1 / churn)



EXAMPLE

- Total spend: $20,000
- New customers: 500
- **CAC** = $20,000 / 500 = **$40 per customer**
- ARPU = $10 / month
- Gross Margin = 80%
- Avg. Lifetime = 12 months
- **LTV** = 10×0.8×12 = **$96 per customer**

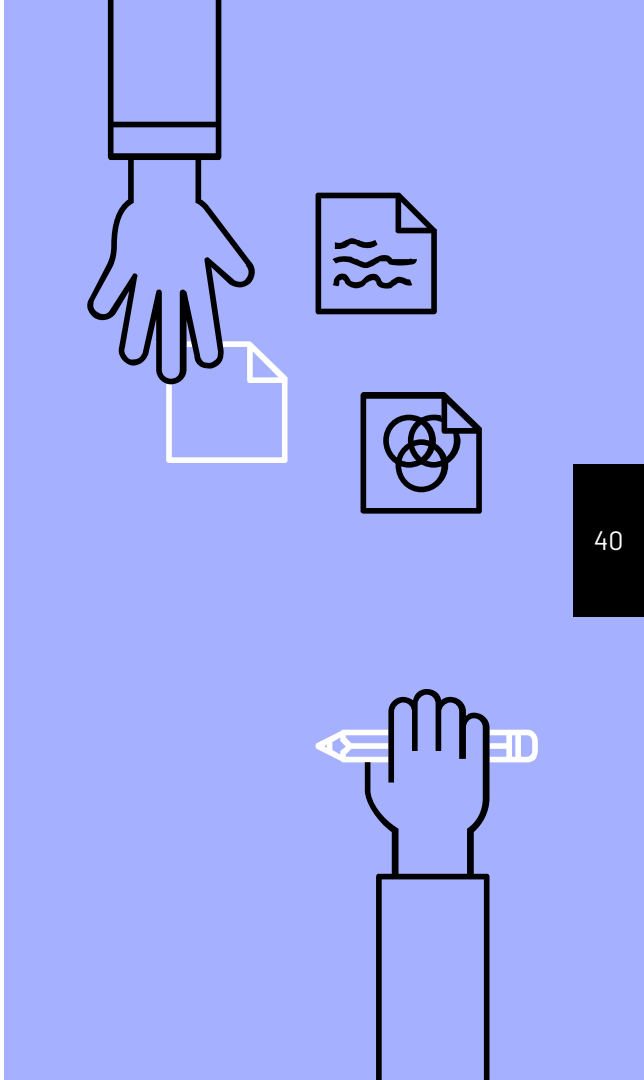
- **Ratio LTV : CAC** = 96/20 = **2.4**

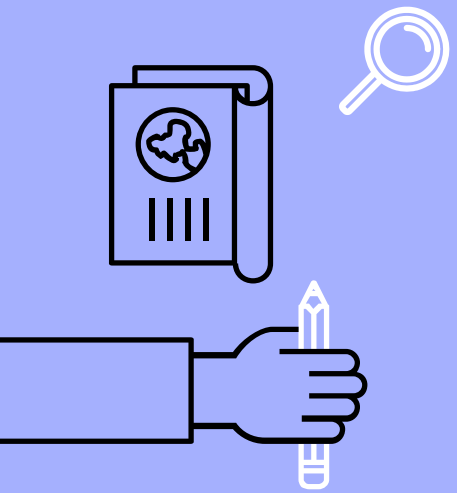


LTV : CAC RATIO

- A healthy business typically maintains **LTV ≥ 3 × CAC**
- If **LTV:CAC < 1**, you're losing money on each user.

Would you rather have a low CAC or a high LTV — and why?





Fix the Unit Economics

Which lever to improve first — CAC, Retention, or Pricing?

	Startup A	Startup B	Startup C
CAC	60€	25€	100€
ARPU	12€/month	8€/month	25€/month
Churn	10%	20%	5%
LTV	120€	40€	500€
LTV : CAC	2 : 1	1.6 : 1	5 : 1

Churn & Retention

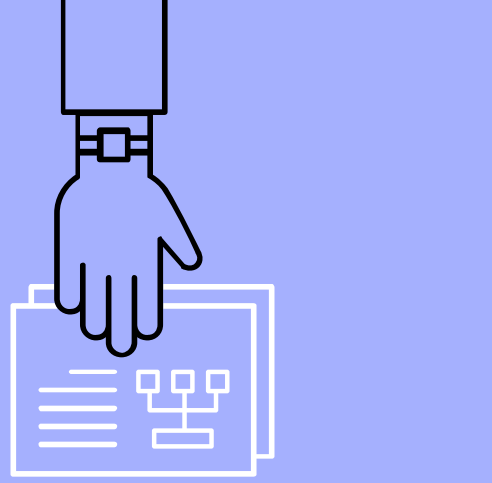
$$Churn = \frac{\text{Customers at Start} - \text{Customers at End}}{\text{Customers at Start}}$$

Retention = 1 - Churn

- Example:
  - Start: 1,000 customers
  - End: 850 customers

→ Churn = (1000 - 850) / 1000 = 15%

→ Retention = 85%

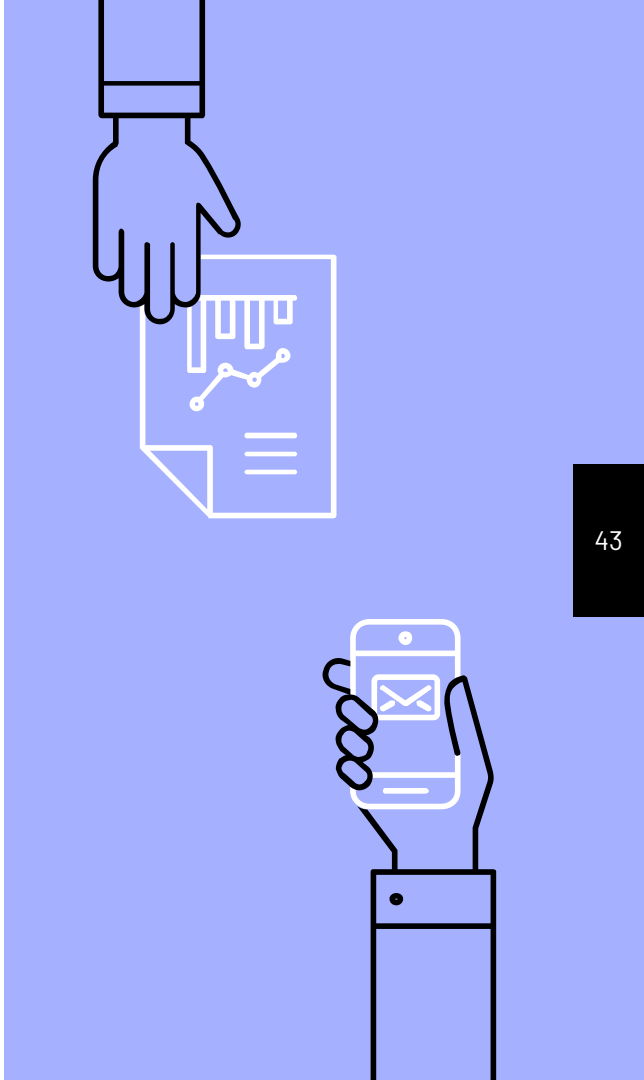




COHORT ANALYSIS

Group users by signup date (cohort) and track retention over time.

Signup month	Month 1	Month 2	Month 3	Month 4
January	100%	80%	60%	45%
February	100%	85%	70%	50%
March	100%	90%	75%	60%

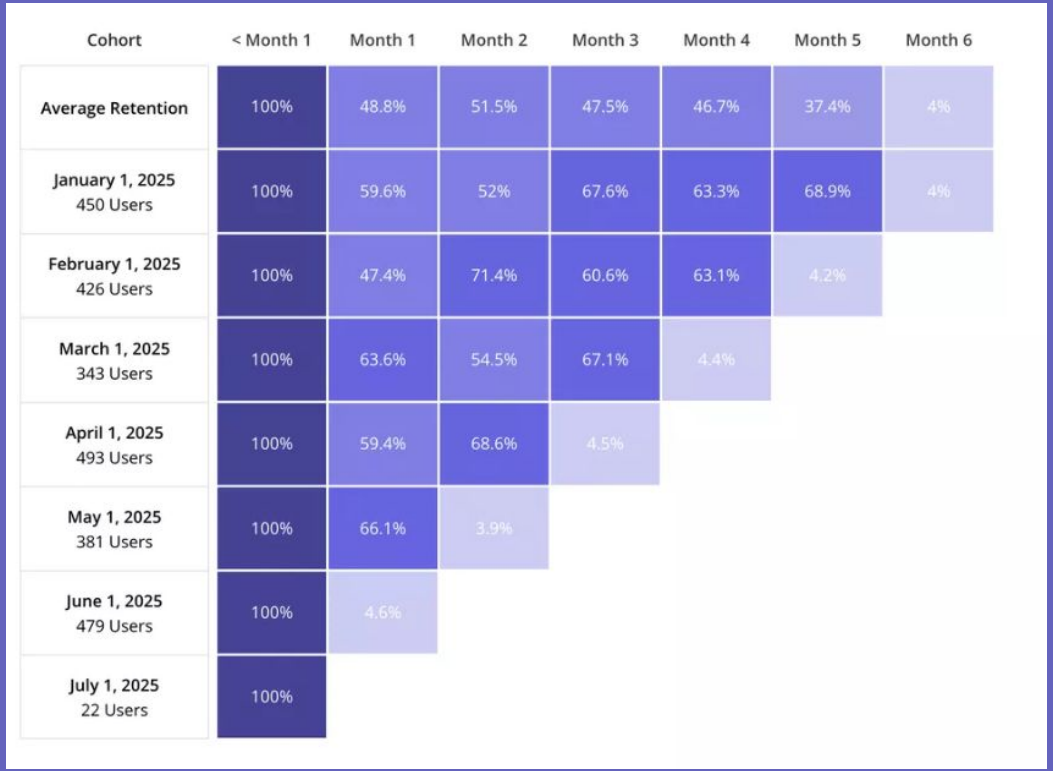


Retention improving over time

→ product improvements or better onboarding.

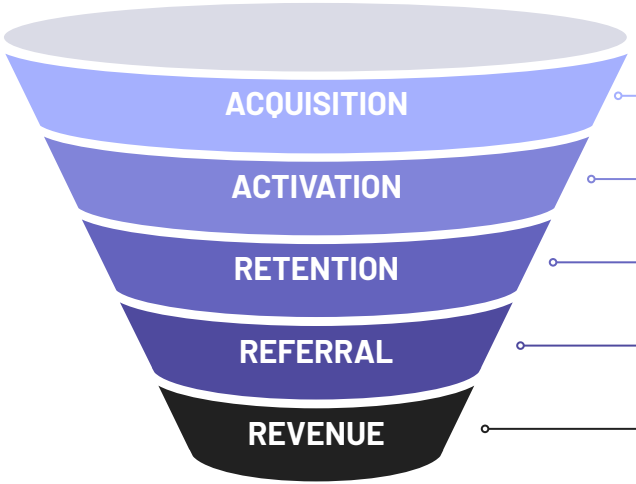
Retention declining

→ user fatigue or poor engagement.



Cohort	< Month 1	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Average Retention	100%	48.8%	51.5%	47.5%	46.7%	37.4%	4%
January 1, 2025 450 Users	100%	59.6%	52%	67.6%	63.3%	68.9%	4%
February 1, 2025 426 Users	100%	47.4%	71.4%	60.6%	63.1%	4.2%	
March 1, 2025 343 Users	100%	63.6%	54.5%	67.1%	4.4%		
April 1, 2025 493 Users	100%	59.4%	68.6%	4.5%			
May 1, 2025 381 Users	100%	66.1%	3.9%				
June 1, 2025 479 Users	100%	4.6%					
July 1, 2025 22 Users	100%						

FUNNEL AARRR



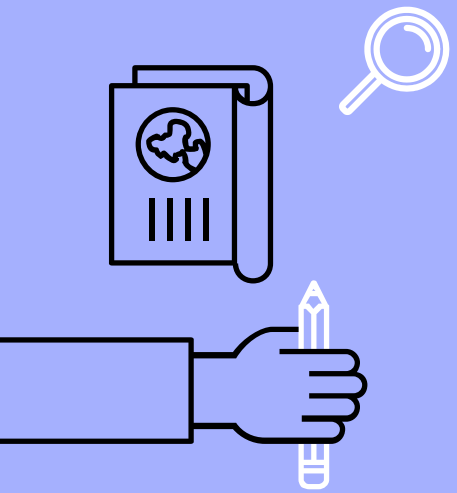
attracting - **CAC (cost of acquisition)**

engaging - activation rate, time to conversion

retaining - **Churn/Retention**

promoting - NPS

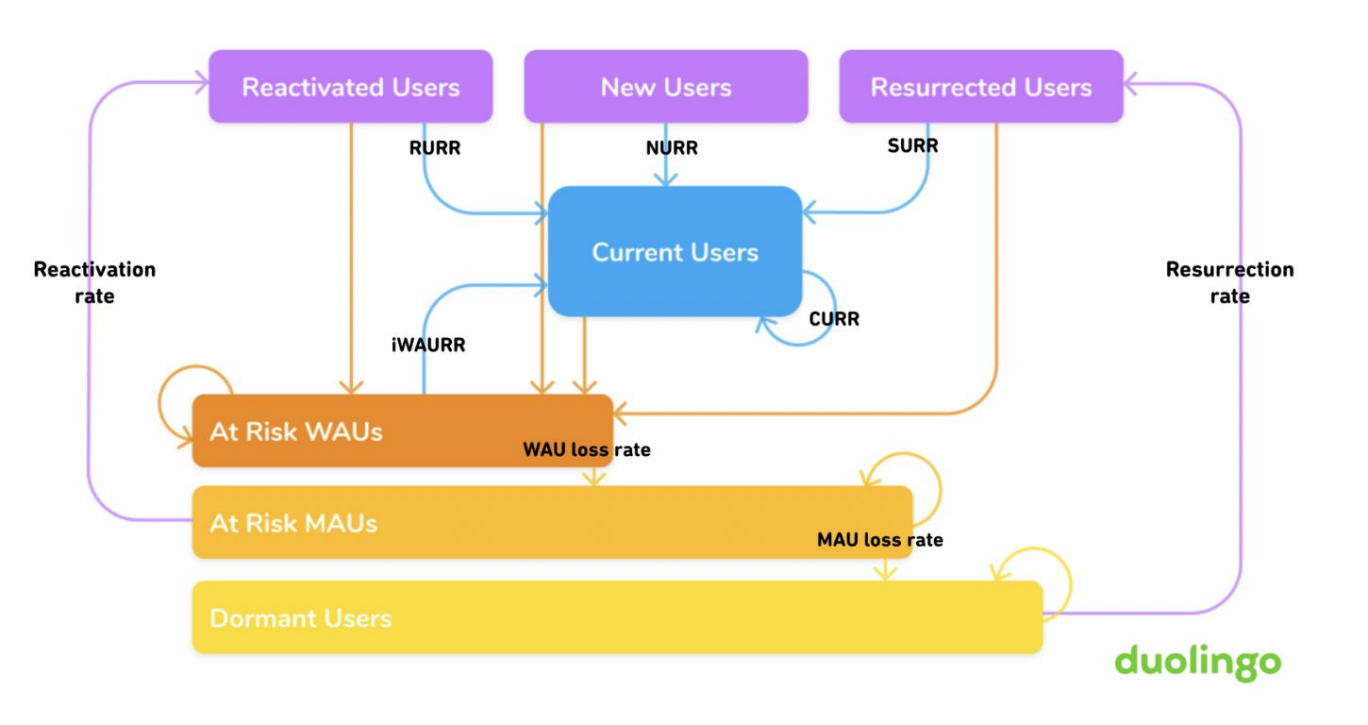
monetising - **LTV (Lifetime Value)**, ARPU



Cohort exercise

- You start with 1,000 users.
- Monthly churn at 10%.
- What happens if you reduce churn by just 2 percentage points (from 10% → 8%)?
- How many more users stay after 6 months?

DUOLINGO EXAMPLE



Virality & Word of Mouth

How many new users each existing user brings in?

$$K = i \times c$$

- **i** = invites per user
- **c** = conversion rate

- Example :
  - 1000 users
  - each user invites 2 friends → i = 2
  - 30 % of those friends sign up → c = 0.3
  - Then: K = 2 × 0.3 = 0.6

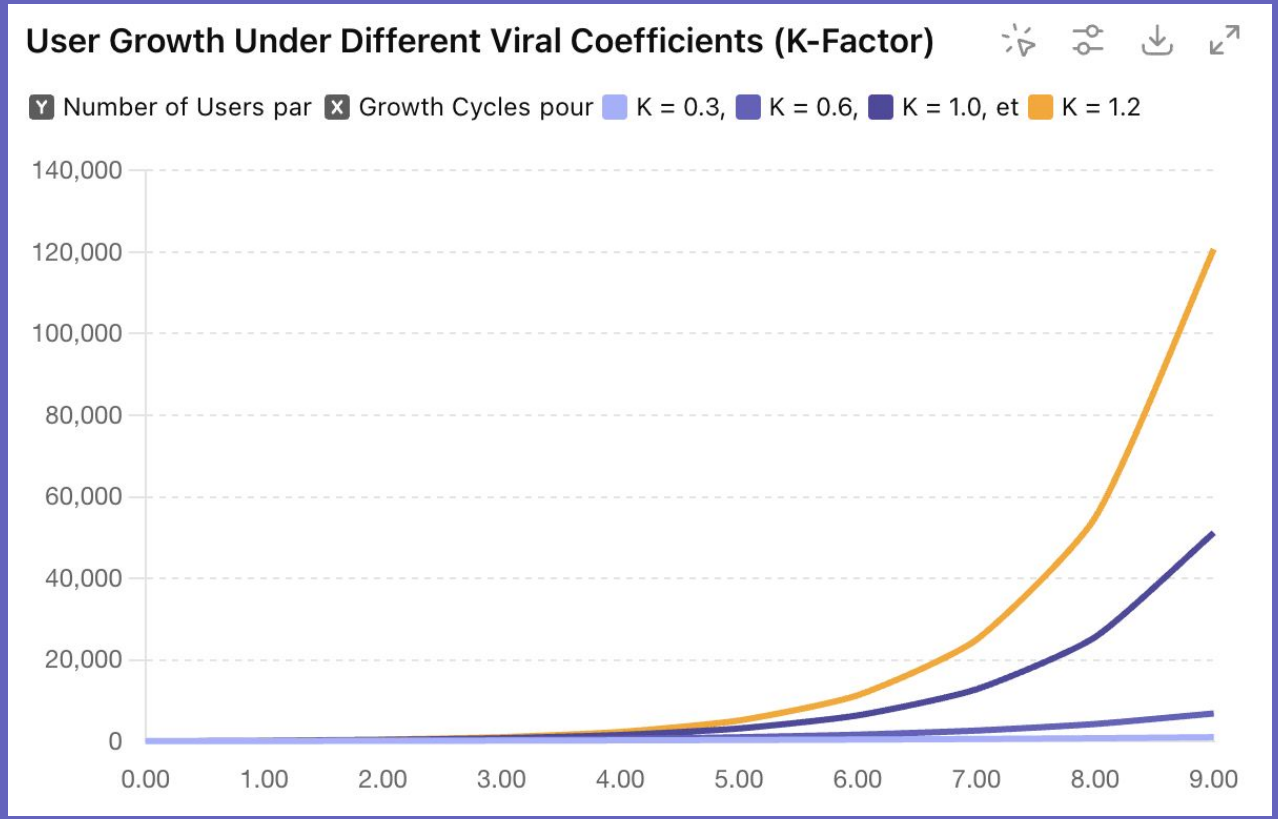
→ every user generates 0.6 new users on average



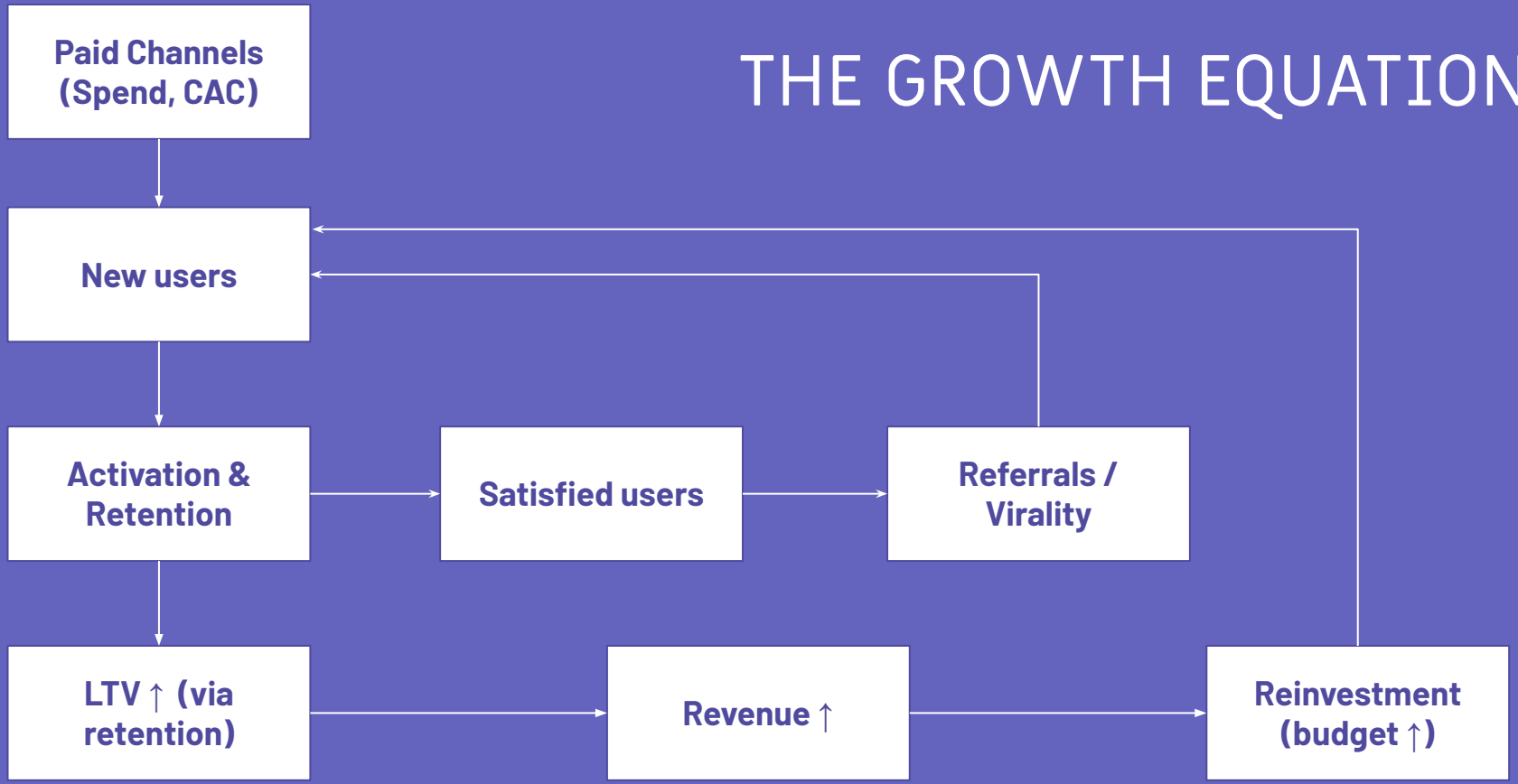
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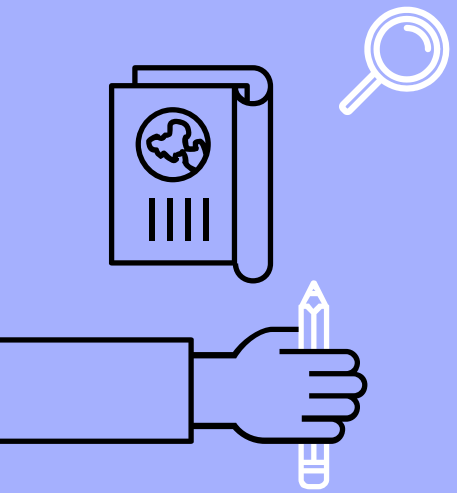
- **K > 1** → the loop fuels itself (viral growth engine).
- **K < 1** → the loop helps, but doesn't replace acquisition.
- **K ≈ 0.3–0.8** → realistic sweet spot; good products use it to stretch their marketing efficiency, not eliminate it. Not viral alone but multiplies ads efficiency.




User Growth Under Different Viral Coefficients (K-Factor)
Y Number of Users par
X Growth Cycles pour
K = 0.3,
K = 0.6,
K = 1.0, et
K = 1.2
140,000
120,000
100,000
80,000
60,000
40,000
20,000
0
0.00
1.00
2.00
3.00
4.00
5.00
6.00
7.00
8.00
9.00

THE GROWTH EQUATION



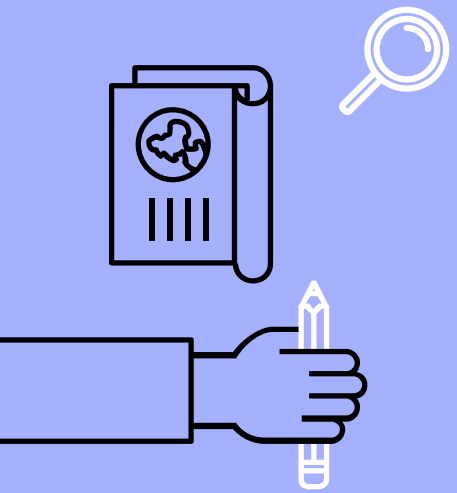


Time for a quizz (1/3)

A startup spends €40 000 to acquire 1 000 new paying users. Each user pays €20/month and churns at 10 % per month.

Which statement is true?

A. CAC = €20; LTV ≈ €200 → healthy unit economics
B. CAC = €40; LTV ≈ €200 → borderline sustainable
C. CAC = €40; LTV ≈ €100 → unsustainable
D. CAC = €20; LTV ≈ €100 → strong growth loop

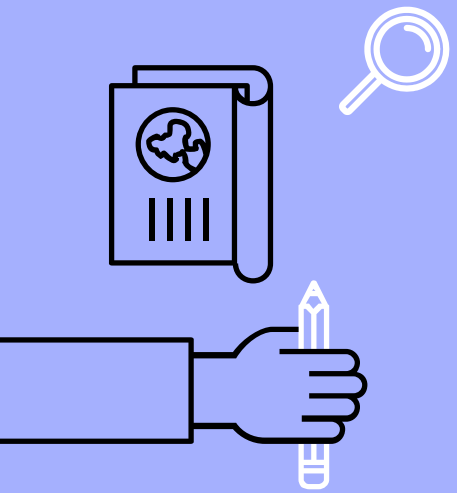


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Time for a quizz (2/3)

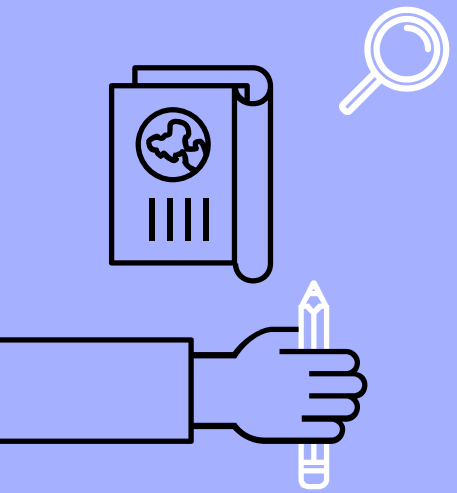
If a product improves its monthly churn from 10 % to 5 %, what happens to its LTV, assuming ARPU = €10?

A. LTV stays the same — churn doesn't affect value.

B. LTV decreases — fewer users churn means less new acquisition.

C. LTV increases from €100 to €150.

D. LTV doubles from €100 to €200.



Time for a quizz (2/3)

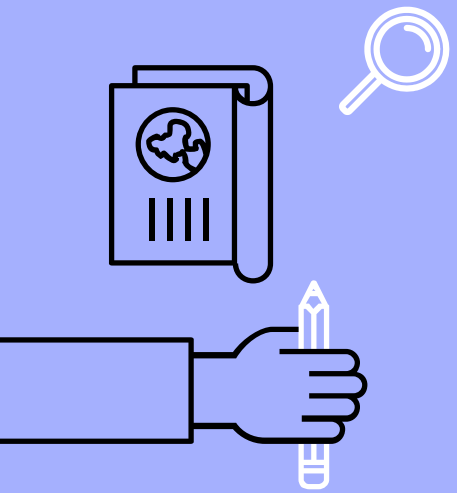
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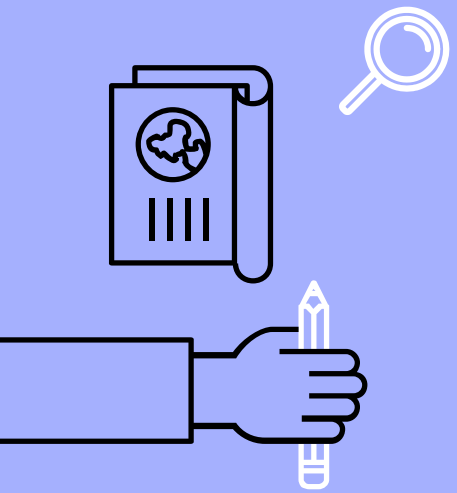
D. LTV doubles from €100 to €200.



Time for a quizz (3/3)

An app's users send an average of **2 invites** each, and **25 %** of invitees join. What's the app's viral coefficient (K-factor), and will it grow exponentially?

A. K = 0.25 ⟶ no, it'll decay
B. K = 0.5 ⟶ no, it needs fuel
C. K = 1 ⟶ yes, it's self-sustaining
D. K = 2 ⟶ yes, infinite growth



Time for a quizz (3/3)

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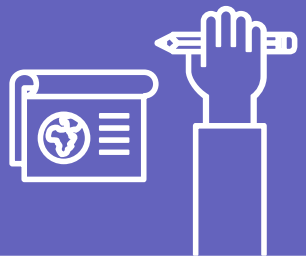
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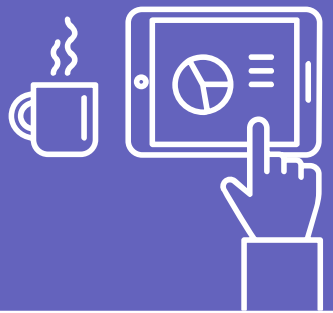
Which metric is most dangerous to ignore?

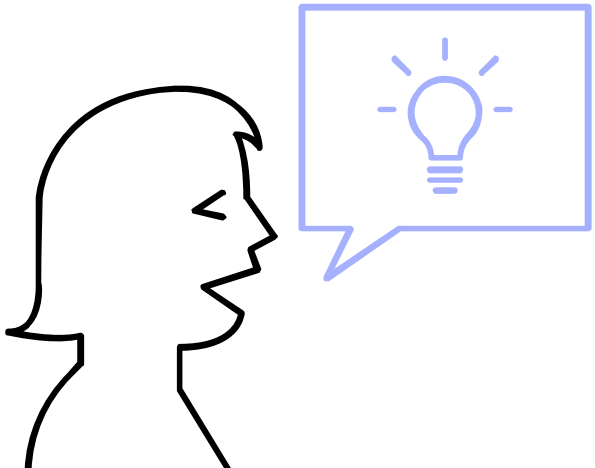


Why can a low CAC be a bad sign?

How does retention signal product market fit?







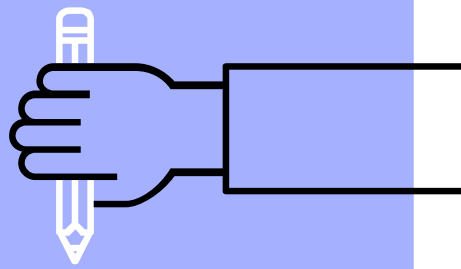
CAC:LTV ratio determines scalability

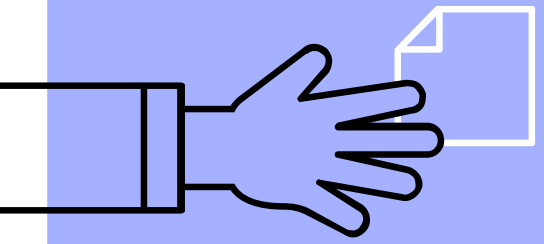
Retention compounds faster than acquisition.

Virality amplifies retention.

Improving retention is usually the smartest growth lever.

3.
ACQUISITION & THE ETHICS OF GROWTH





Not all growth is created equal.

Paid Channels

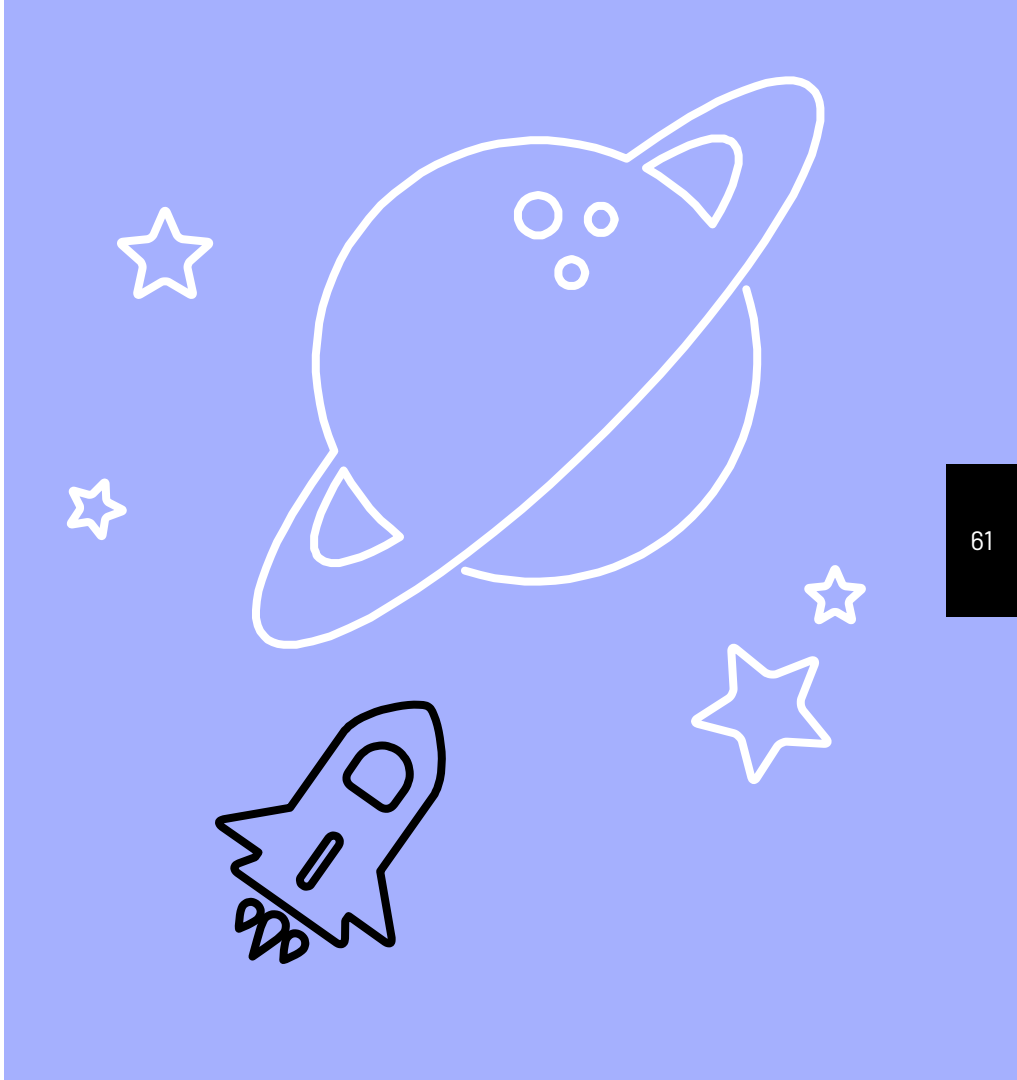
Fast but costly

Organic Channels

Slow but durable

Referrals & Incentives

Dark Patterns & Ethics



WHERE DID YOU HEAR ABOUT IT?

Think of the last 3 apps or products you started using.

How did you discover them?



“

Growth

isn’t one channel;

it’s a portfolio.



THE ACQUISITION SPECTRUM

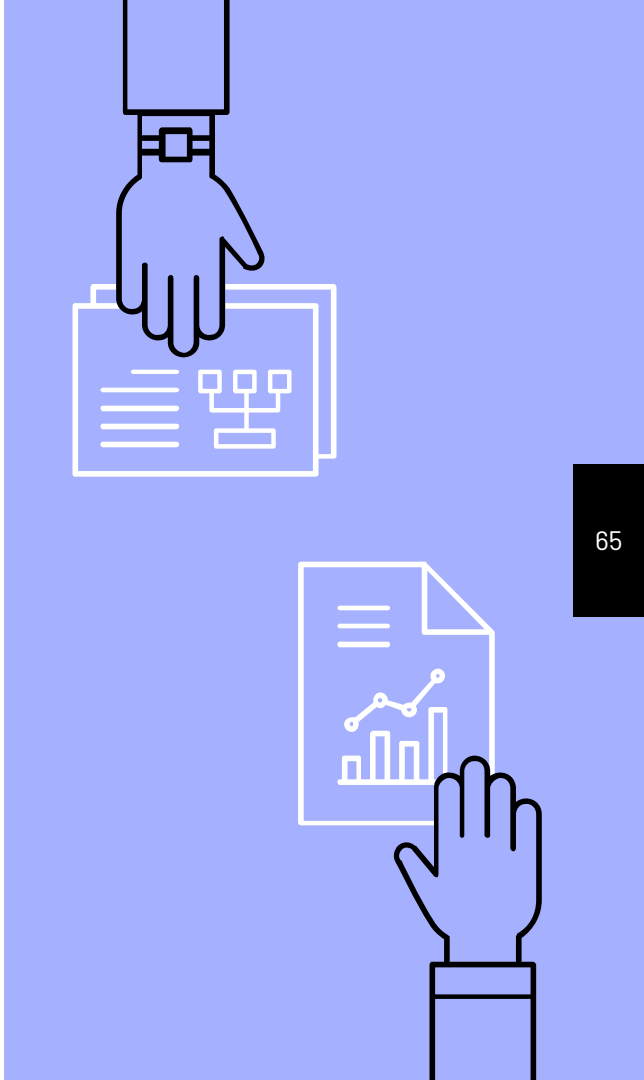
Paid	Organic
• Ads • Influencers • Affiliates	• SEO • Community • Referrals

PAID CHANNELS: THE FAST LANE

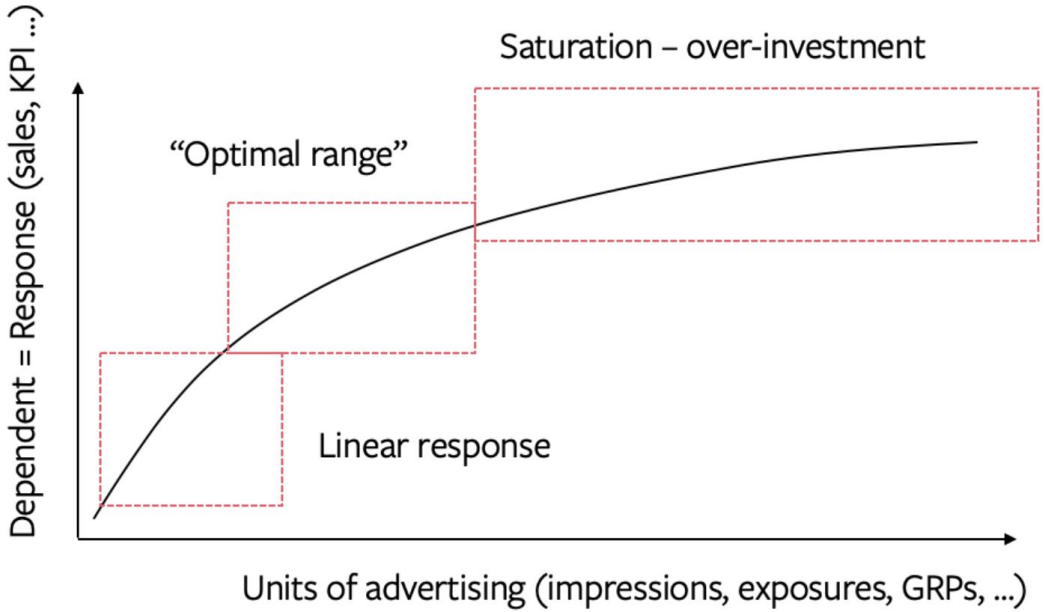
Paying for attention

- **Search** (Google Ads) - Intent driven
- **Social** (Meta, TikTok) - Interest driven
- **Display** - Interest driven
- Cost metrics: CPM, CPC, CPA

$$CAC = \frac{CPC}{ConversionRate}$$

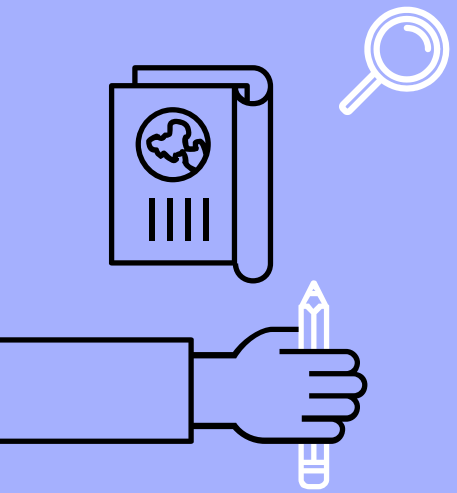


PAID CHANNELS DYNAMICS



After a certain point, also called the saturation point, each dollar you spend has a relatively lower impact.

" Why does doubling ad budget often not double users?"



Paid Channel exercise

- Startup spends €20k/month on ads
- CAC = €50
- churn = 10%
- You get +10 % budget

- Do you spend, optimize, or pivot?

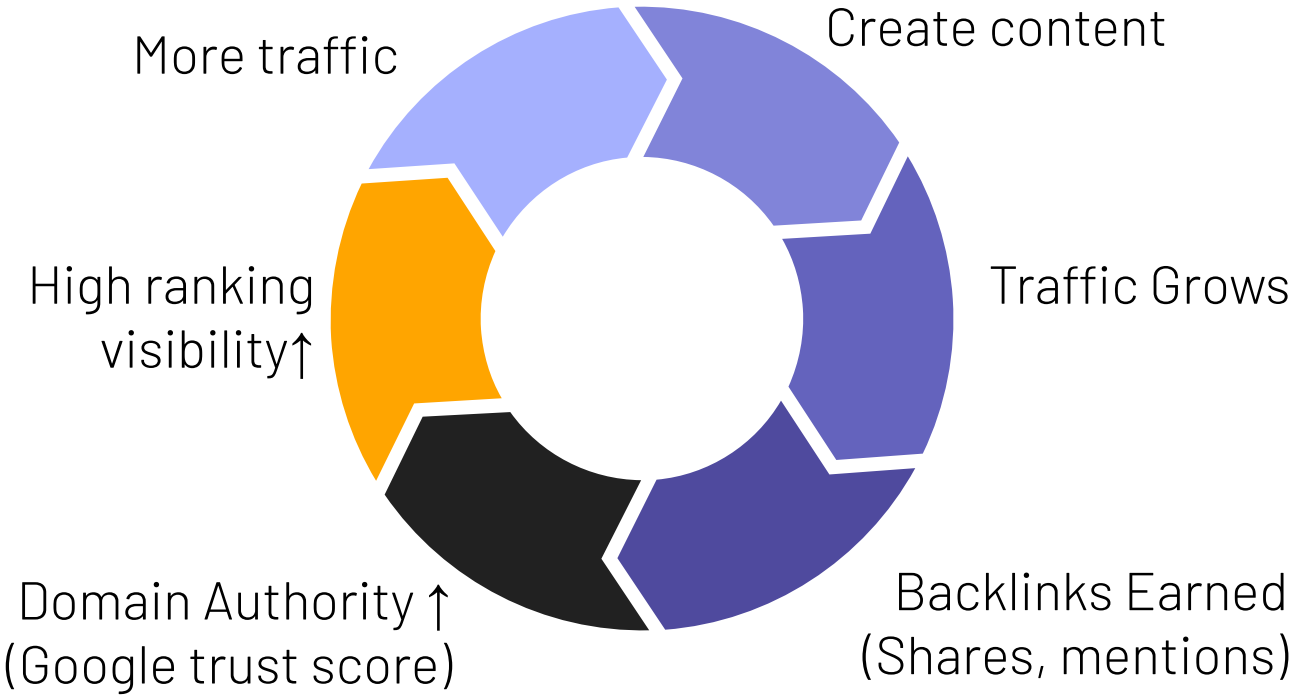
ORGANIC CHANNELS: THE LONG GAME

Building trust

- SEO
  - search-based, high intent, long setup time.
- Content Marketing
  - builds authority and brand.
- Community & Product-Led Growth
  - advocacy, word of mouth through product value.
- Referrals & Virality:
  - built-in acquisition loops.



THE SEO FLYWHEEL



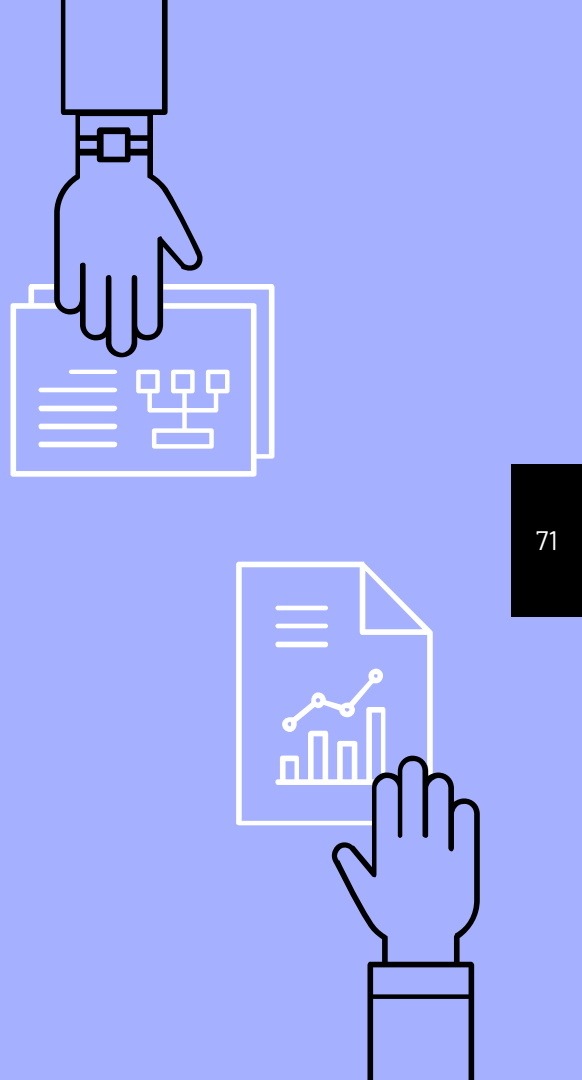
PAID vs ORGANIC TRADEOFFS

	🕑 Fast Impact	⏳ Slow Impact
💸 **Paid**	**Ads, Influencers, Affiliates** - Instant reach - Scalable with budget - Diminishing returns (rising CAC)	**Partnerships, Sponsorships** - Brand alignment builds credibility - Slow setup, but durable reach
🌱 **Organic**	**Referrals, Social Virality** - Unpredictable spikes - Hard to control, low cost - Strong trust signal	**SEO, Content, Community** - Compounding traffic - High upfront effort, near-zero CAC long term

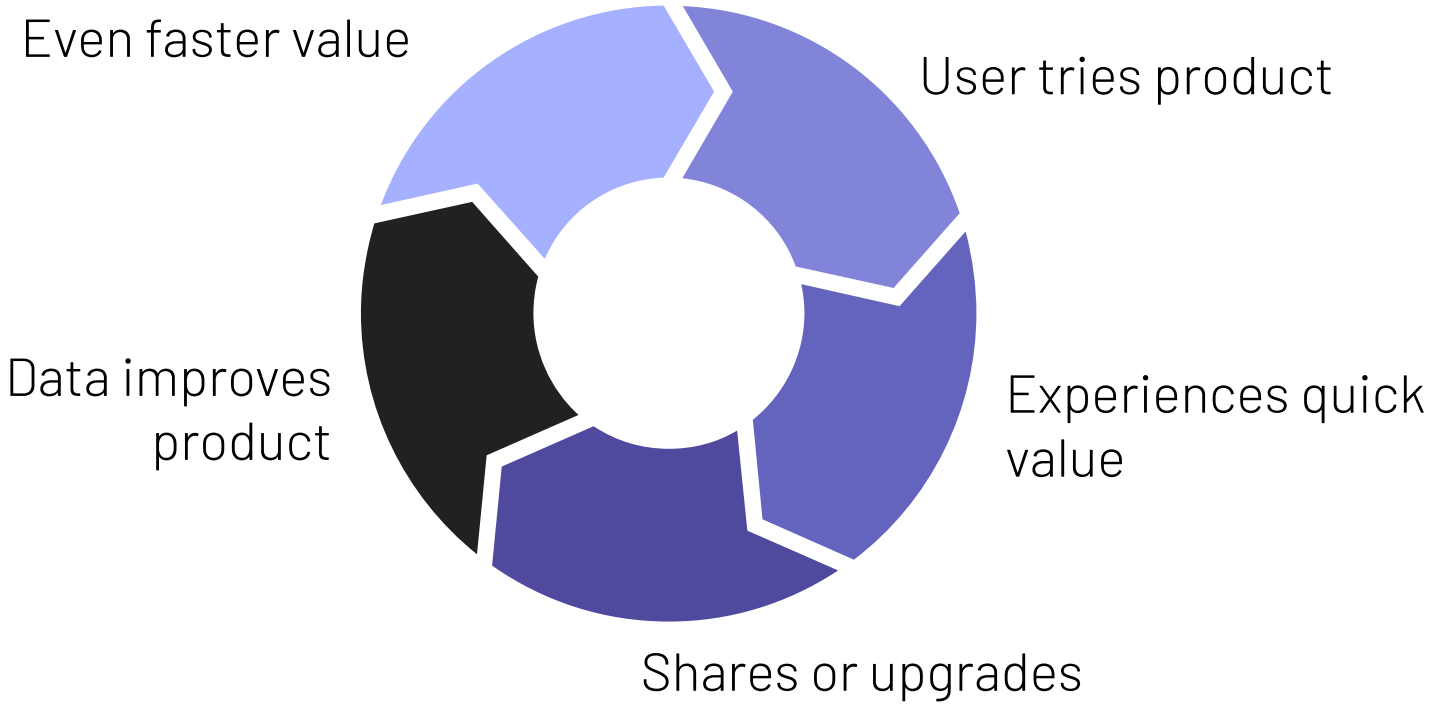
PRODUCT LED GROWTH

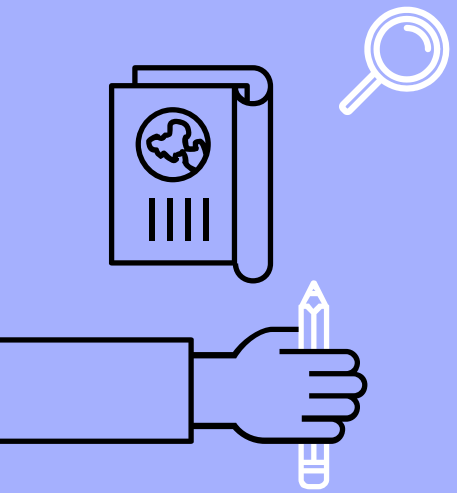
A go-to-market strategy where the product itself drives acquisition, activation, and retention.

- **The product = the main growth engine**
  Instead of relying primarily on sales or marketing
- Users **experience value**
  they convert, invite others, and expand usage.
- The **feedback loop** between product usage and business metrics is immediate and measurable



THE PRODUCT FLYWHEEL





PRODUCT EXERCISE

- Find a company that has a strong product
- Analyse the product and identify how its design leads to growth
- Examples : Slack, Notion, Trello,...

REFERRALS & INCENTIVES

Self-reinforcing growth mechanism where each new user can directly create more users — if the incentive and experience align

- **The mechanics:**

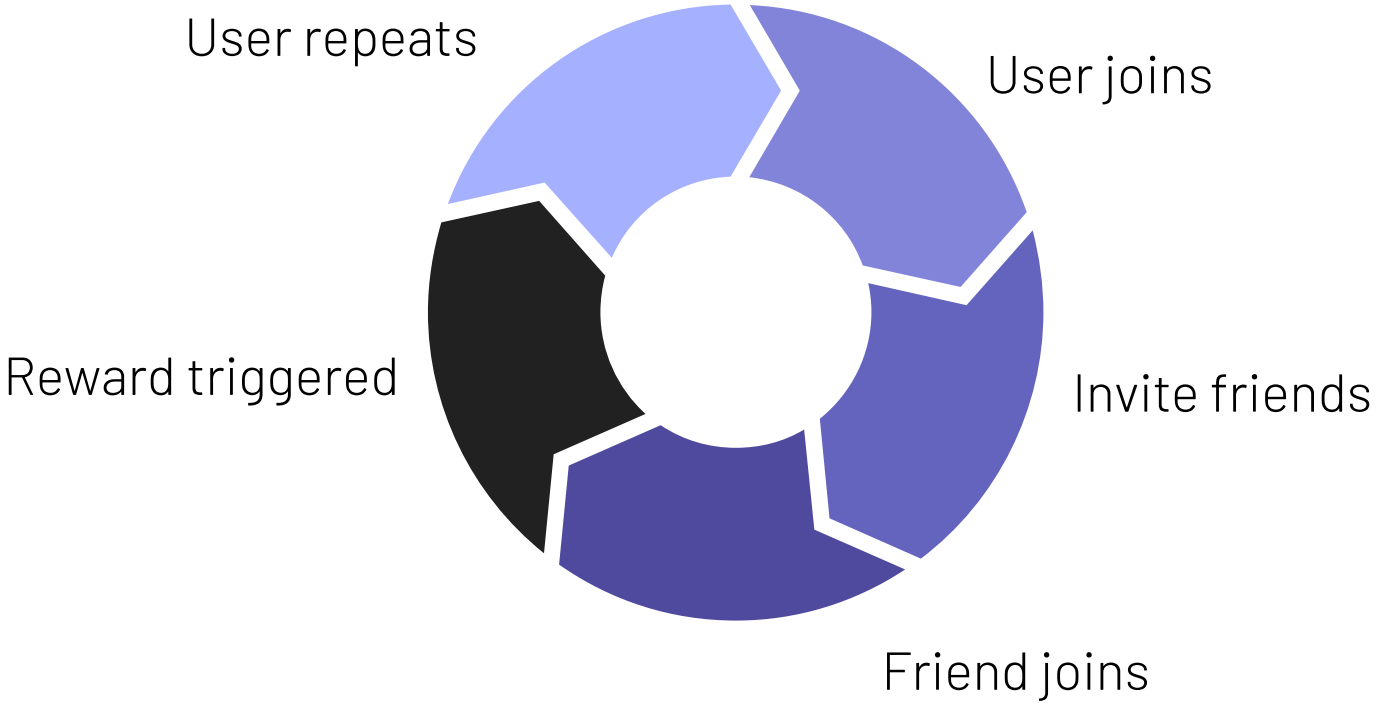
  Trigger → Incentive → Simplicity.
- **Behavioral loop**

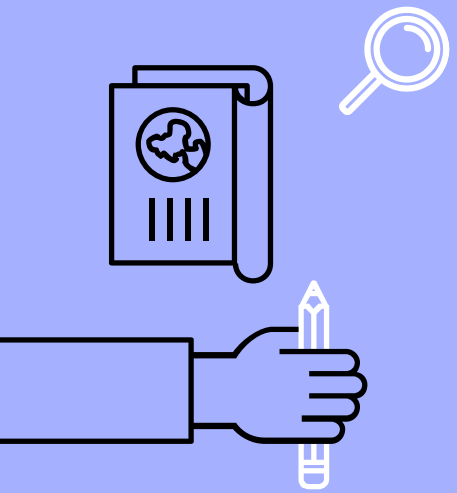
  that depends on motivation, trust, and design.
- Viral coefficient K = i x c
- The **reward structure** determines how likely users are to refer

  ⚠️ Pitfall: over-incentivization → low-quality users.



THE REFERRAL FLYWHEEL





REFERRAL EXERCISE

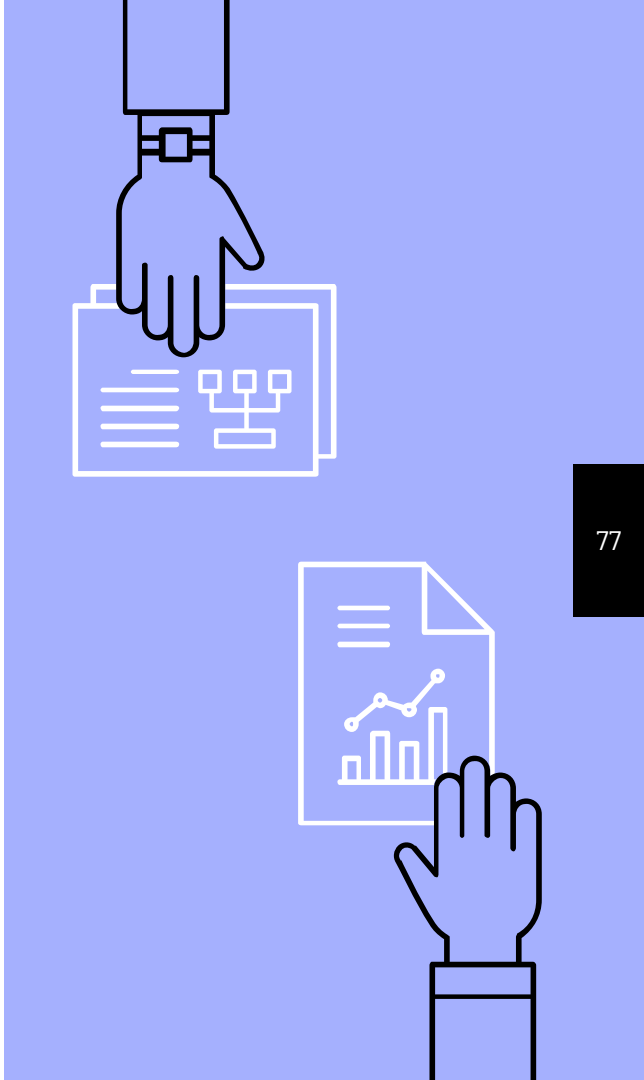
- You’re designing a referral loop for a language-learning app. You can reward users with:
  - €5 cash
  - Free premium lessons
  - Public leaderboard points

- Which do you choose, and why?

DARK PATTERNS

When growth crosses the line

- Hidden unsubscribe buttons
- Guilt-shaming copy (“Are you sure you want to miss this deal?”)
- Auto-opt-ins / forced sharing

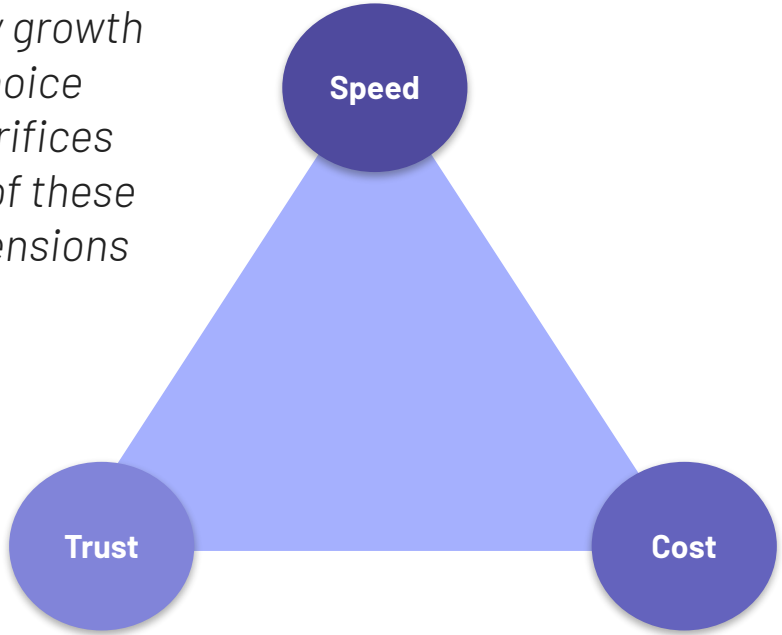


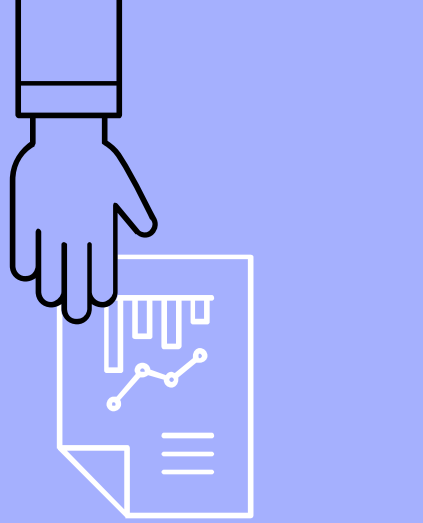
DARK PATTERNS EXAMPLES

Company	Pattern Type	Short-Term Win	Long-Term Risk
LinkedIn	Forced virality	Fast user growth	Lawsuits, user backlash
Facebook	Data opacity	Higher engagement	Regulatory scrutiny
Amazon	Obstruction	Lower churn	Loss of brand goodwill
Instagram	Addictive loop	High retention	Mental health & PR issues
Duolingo	Emotional triggers	Daily actives ↑	Ethical criticism
Ryanair	Hidden costs	Higher ARPU	Customer distrust

BALANCING GROWTH FORCES

Every growth choice sacrifices one of these dimensions

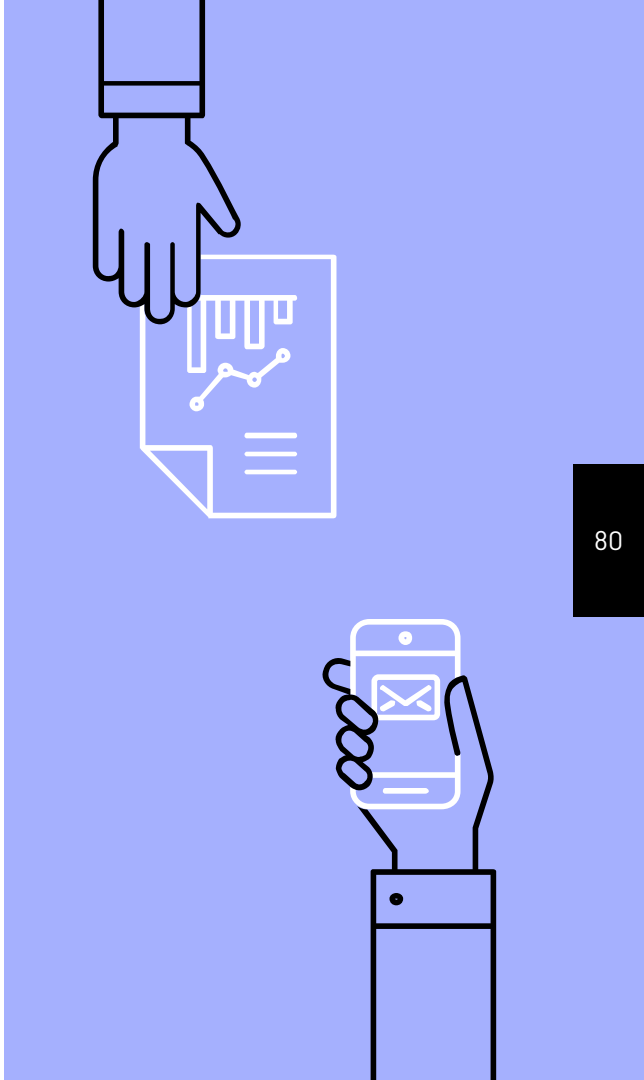






EXAMPLES

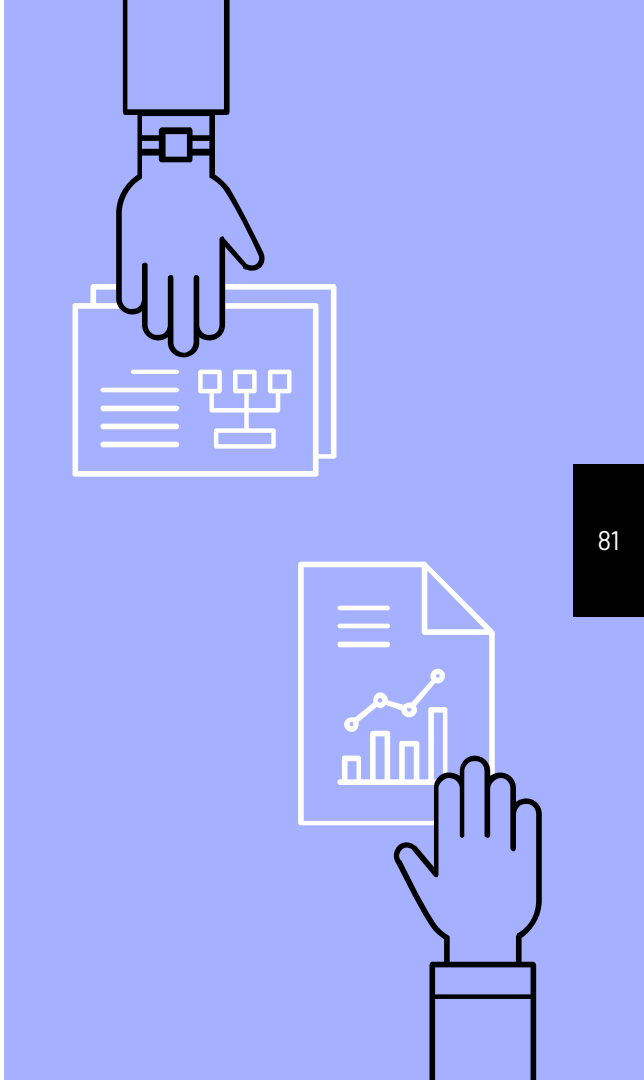
Growth Type	Strength	What you sacrifice	Strategic Use
Paid	Speed	Cost & Trust	Early acceleration, testing, launching
Organic	Trust	Speed & Control	Sustainable base, brand moat
Referral / Virality	Balance (efficiency + Credibility)	Predictability & Control	Mid-stage multiplier, network effects

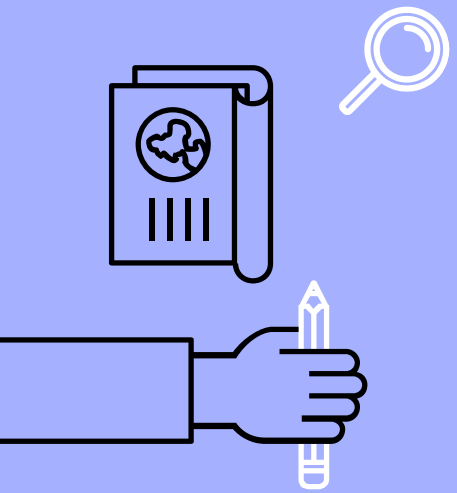


Ethical Growth Principles

Sustainable growth is not anti-growth — it's long-term optimization

- Align incentives with **user value**
- **Be transparent** about trade-offs
- Optimize for **lifetime trust**, not one-time clicks



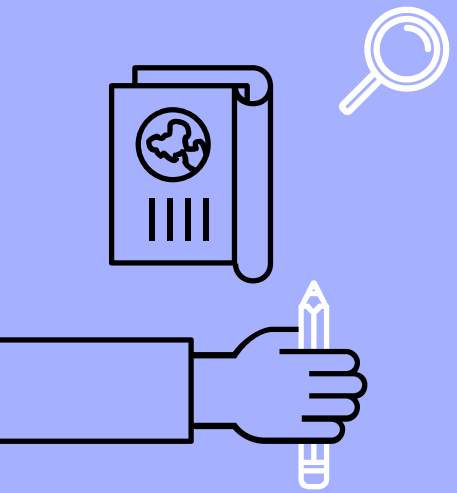


Time for a quizz (1/3)

A new app claims a “K-factor” of 1.2 because each user sends 4 invites and 30 % convert. After launch, they discover most new users come from paid ads, not invites.

What’s the real issue?

A. They confused potential virality with effective virality.
B. Their CAC is too high.
C. They need more incentive per invite.
D. Their K-factor formula is wrong.



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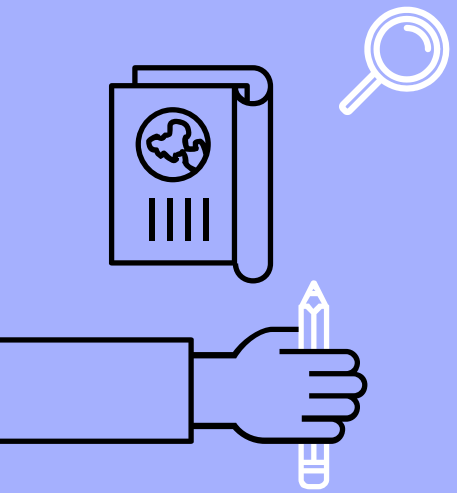
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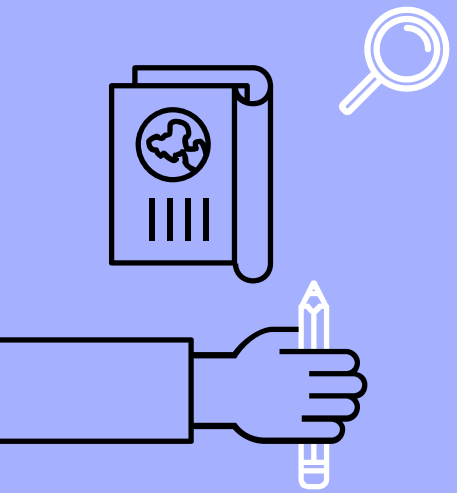


Time for a quizz (2/3)

You run growth for a subscription app. Paid ads deliver users at €40 CAC; SEO delivers at €15 CAC but grows slowly.

What's your best strategic move?

A. Double down on ads for faster growth.
B. Pause SEO — it's too slow.
C. Balance: fund ads to keep growth while compounding SEO for long-term CAC efficiency.
D. Ignore both and launch a referral program.

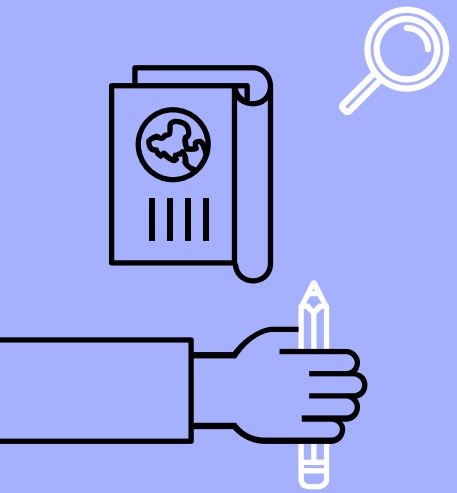


Time for a quizz (2/3)

You run growth for a subscription app. Paid ads deliver users at €40 CAC; SEO delivers at €15 CAC but grows slowly.

What's your best strategic move?

A. Double down on ads for faster growth.
B. Pause SEO — it's too slow.
C. Balance: fund ads to keep growth while compounding SEO for long-term CAC efficiency.
D. Ignore both and launch a referral program.

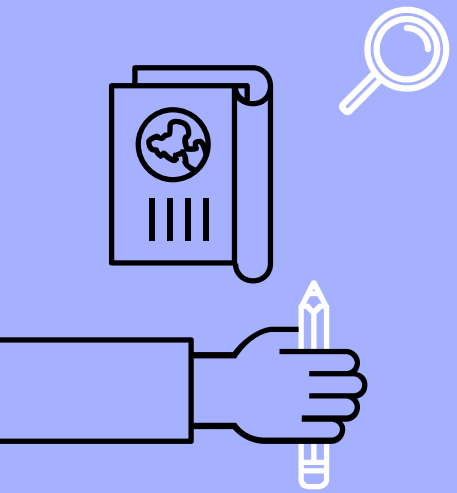


Time for a quizz (3/3)

A mobile game increases retention by adding surprise pop-ups that exploit “fear of missing out.” Daily actives rise 30 %, but user complaints explode.

What’s happening?

A. Smart growth: retention up = success.
B. Temporary gain: dark pattern damages trust and long-term retention.
C. Irrelevant — engagement is the only KPI.
D. Needs better A/B testing.



Time for a quizz (3/3)

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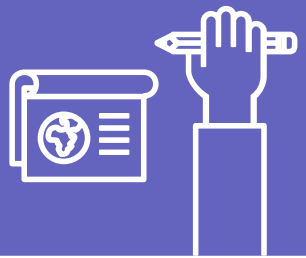
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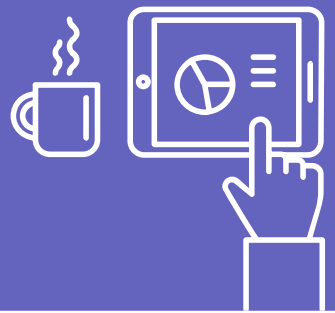
Which channel builds the strongest moat?

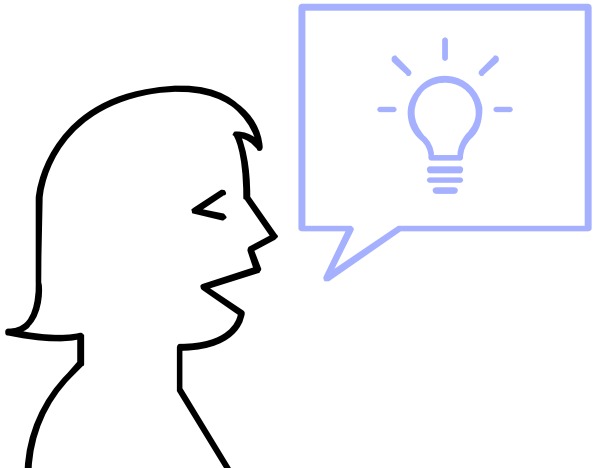


Can growth be 100 % ethical?

Where would you draw the moral line if you ran TikTok's growth team?

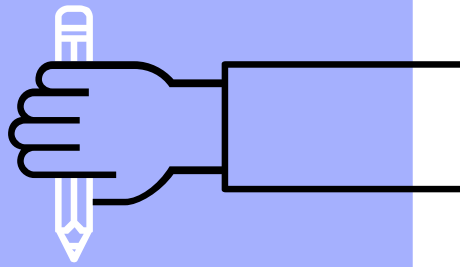


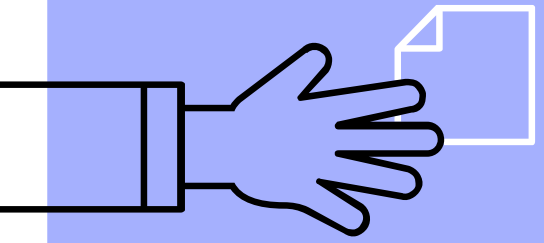




- **Paid fuels speed, organic builds the moat**
- **Referrals & product loops scale trust : user become your best acquisition channel**
- **Dark patterns break compounding**

4.
MODELING GROWTH & SCENARIO BUILDING





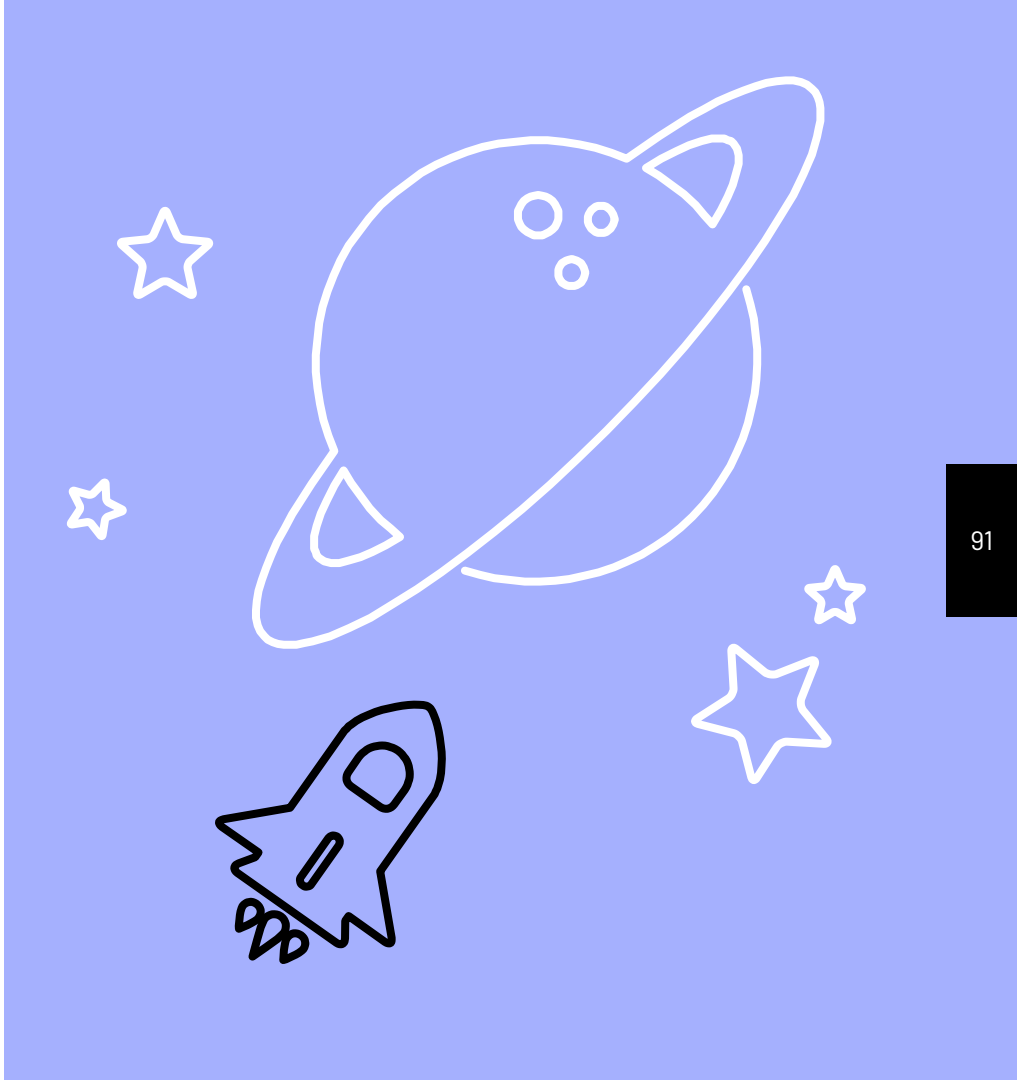
Turning growth from gut feeling into simulation

Anatomy of a Growth Model

Scenario Planning (Optimistic / Base / Pessimistic)

From Numbers → Strategy

Ethics & Realism



IF YOU HAD 10% MORE BUDGET



you're a Growth Lead at a subscription app

- **10 k users**
- **CAC : €40**
- **ARPU : €12/mo**
- **Churn : 8 %**

Where do you invest?



Why model?

Good growth models don't predict the future — they show how fragile your beliefs are.



💰 B2B SAAS GROWTH MODEL

💡 Adjust the Actuals using the months that best represent the baseline for the upcoming quarter.

	Actuals			Simulation		
	August 2024	September 2024	October 2024	M+1	M+2	M+3
GROSS NEW $ARR	**$327,220**	**$428,334**	**$330,059**	**$361,381**	**$361,381**	**$361,381**
TRIAL $ARR	**$238,211**	**$264,410**	**$219,319**	**$241,421**	**$241,421**	**$241,421**
Trial forms submitted	**1000**	**980**	**1111**	1030	1030	1030
Verified trials	**850**	**843**	**900**	**865**	**865**	**865**
Trial verification rate	85.00%	86.00%	81.00%	84.00%	84.00%	84.00%
New accounts created	**770**	**774**	**811**	**786**	**786**	**786**
Account creation rate	77.00%	79.00%	73.00%	76.33%	76.33%	76.33%
New customers (PT)	**147**	**149**	**160**	**152**	**152**	**152**
Trial conversion rate	14.70%	15.20%	14.40%	14.77%	14.77%	14.77%

ANATOMY OF A GROWTH MODEL



- Spend
- CAC
- Users Acquired

- Activation Rate
- Churn
- Virality
- ARPU

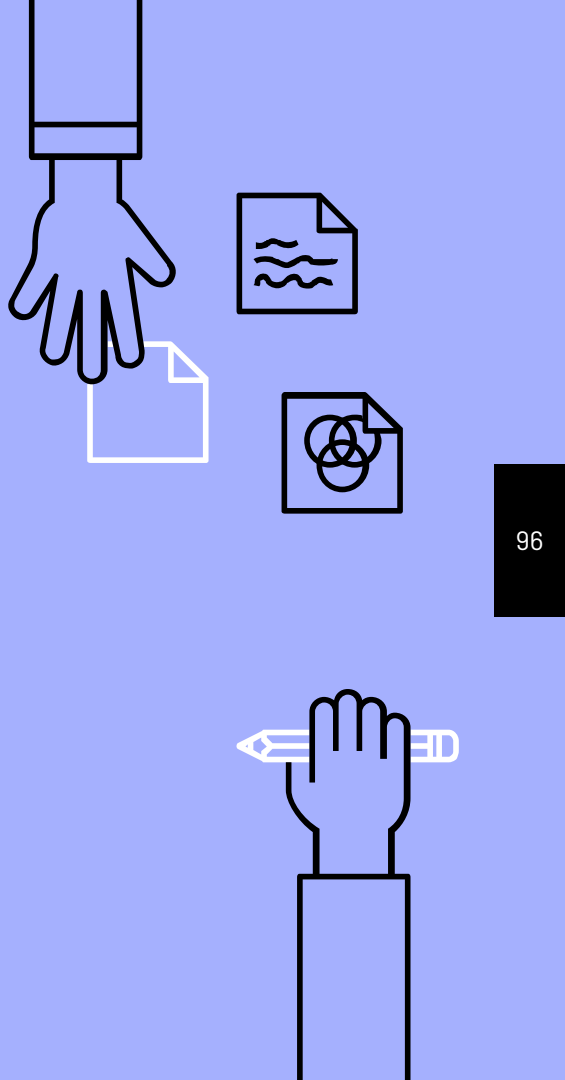
- Revenue
- Active Users
- Growth Rate

$$U_{t+1} = (U_t \times (1 - Churn)) + (NewUsers) + (Referrals)$$

EXAMPLE

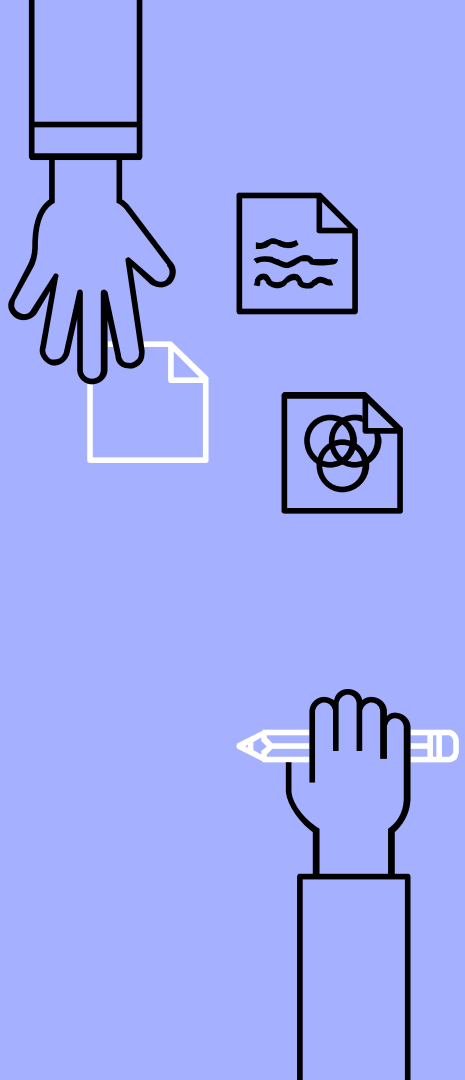
- Start with: **U_0 = 1000 users**
- CAC = €50 → Spend €5000 → 100 new users
- Churn = 10 % → 900 remain
- Virality = 0.2 → 180 referrals
- **U1 = 1 180 next month** → +18 % growth

- What if churn drops to 5 %?
  - 1230
- Virality = 0.4?
  - 1360

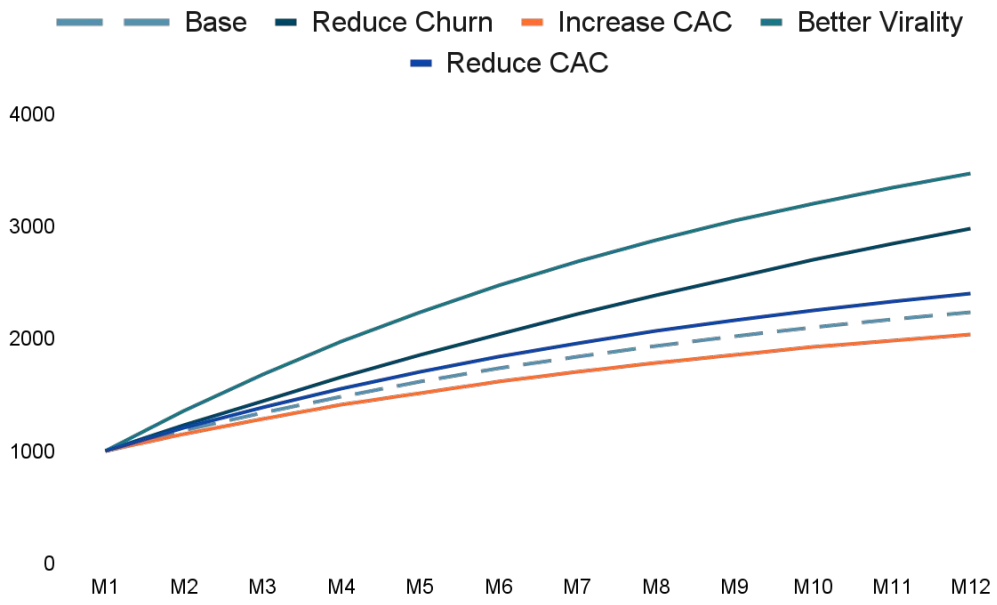


WHAT IF EXPERIMENTS

- Test scenarios as controlled “what if” experiments :
  - **Base case**: current realistic parameters.
  - **Optimistic**: best plausible improvements (virality ↑, Retention ↑).
  - **Pessimistic**: worst-case shocks (e.g., CAC ↑, churn ↑).



WHAT IF EXPERIMENTS



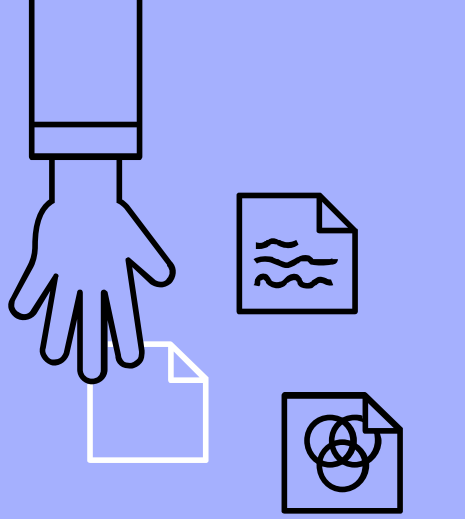
We tweak one parameter at a time

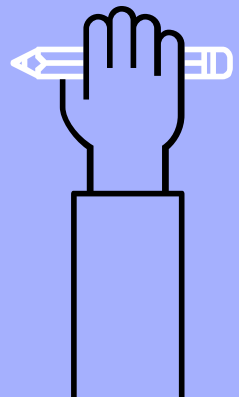
- **churn from 10% to 5%**
- **CAC from 50€ to 70€**
- **CAC from 50€ to 40€**
- **Virality from 0.2 to 0.4**

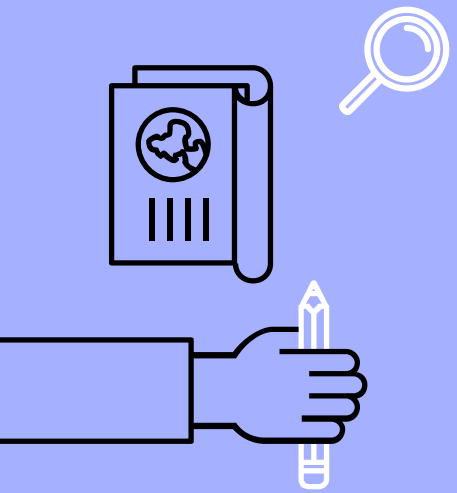
FROM NUMBERS TO STRATEGY

Once you model outcomes, you can derive strategic priorities

Metric	Strategic Lever	Example Action
CAC ↑	Channel Mix	Shift to organic or community
Retention ↓	Product / UX	Improve activation flow
ARPU ↓	Monetization	Upsells or pricing review
K ↓	Network design	Referral simplification







MODELING EXERCISE

Imagine you are the VP of Growth for a B2B SaaS product currently acquiring 1,000 new sign-ups per month. You have an extra $10,000 budget this quarter, and your goal is to maximize Monthly Active Users (MAU) 6 months from now."

You have two strategic options for investment. Which one yields a higher MAU growth, and why?

Scenario 1: Spend $10,000 on product improvements (UX, onboarding).

Scenario 2: Spend $10,000 on paid ads.

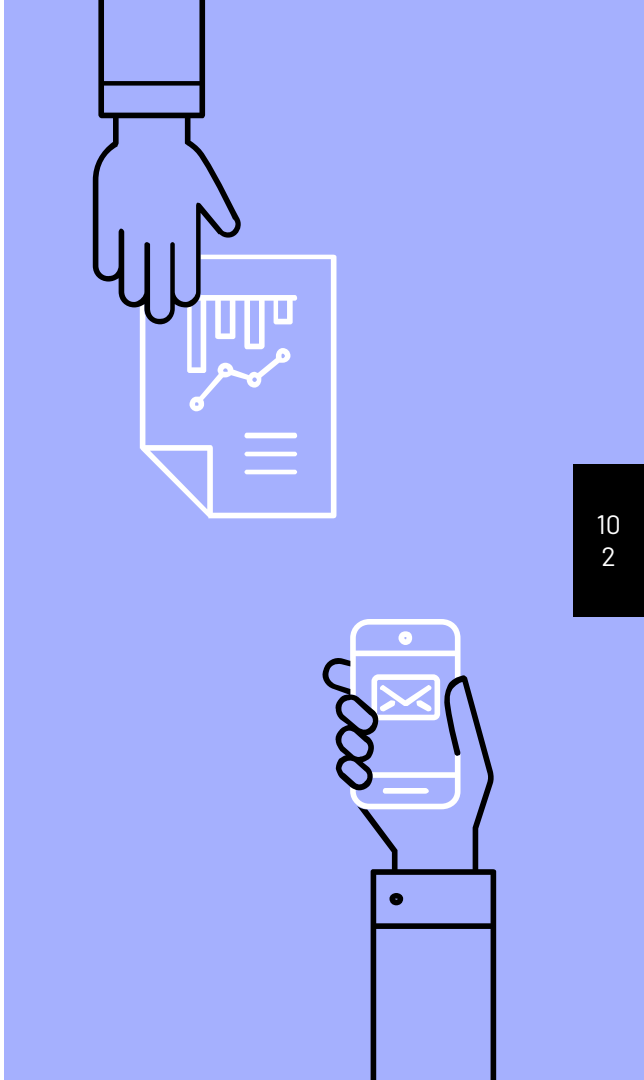
ETHICS & REALISM

If your best scenario depends on doubling screen time, are you creating value or addiction?

MODELING PITFALLS

⚠️ Common errors:

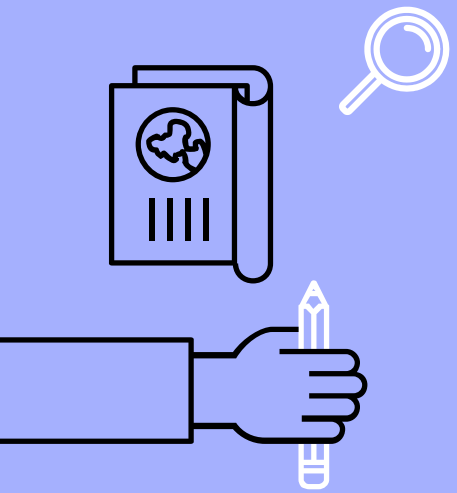
- Assuming linear growth forever.
- Ignoring cost of retention initiatives.
- Confusing virality with word of mouth.



FULL SYSTEM RECAP

Concept	Purpose	Question to ask
Funnel	Diagnose	Where do we leak?
Loop	Sustain	What reinforces itself?
Metrics	Measure	What drives what?
Channels	Execute	Where do we grow?
Model	Integrate	What if we change X?





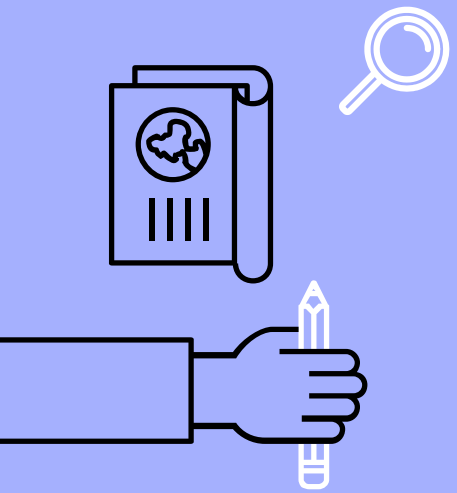
Time for a quizz (1/3)

You build a growth model assuming:

- CAC = €50
- Churn = 5 %
- Monthly ARPU = €25
- Then churn actually rises to 10 %.

What happens to your LTV : CAC ratio?

A. It improves because churn doesn't affect LTV
B. It halves because LTV halves when churn doubles
C. It stays the same because CAC is constant
D. It doubles because higher churn = more frequent purchases



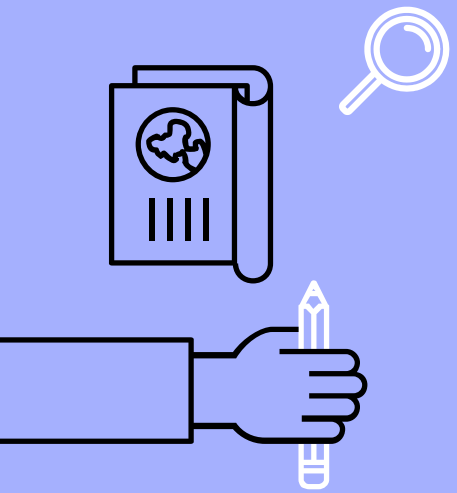
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Time for a quizz (2/3)

Your base model assumes **K = 0.4** (each user brings 0.4 new ones). In an optimistic scenario, **K = 0.8** and churn drops from 8 % → 6 %.

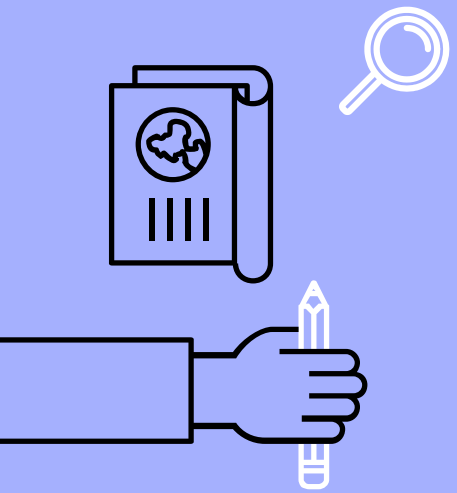
What's the most likely outcome after a few cycles?

A. Growth flattens — virality still < 1

B. Growth accelerates exponentially and becomes self-sustaining

C. Growth doubles instantly in the first cycle

D. Growth slows down because virality reduces CAC



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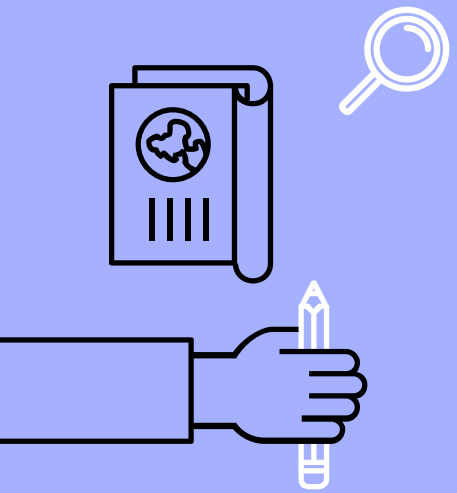
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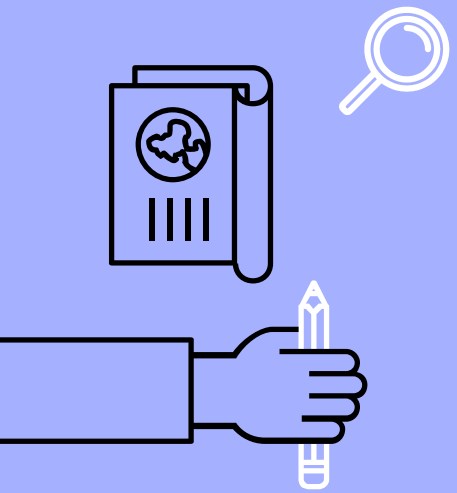
Time for a quizz (3/3)

Your base model:

- CAC = €40
- ARPU = €10/month
- Churn = 10 %

You can improve only one lever by 20 %.
Which gives the biggest impact on LTV : CAC ratio?

A. Reduce CAC by 20 %
B. Increase ARPU by 20 %
C. Reduce churn by 20 % (10 % → 8 %)
D. Doesn't matter — all equal.



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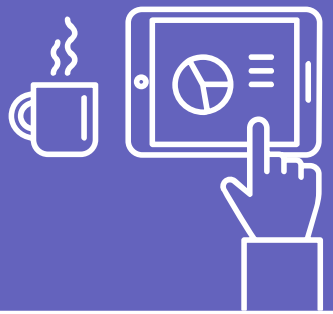
What's the most sensitive assumption in any growth model?

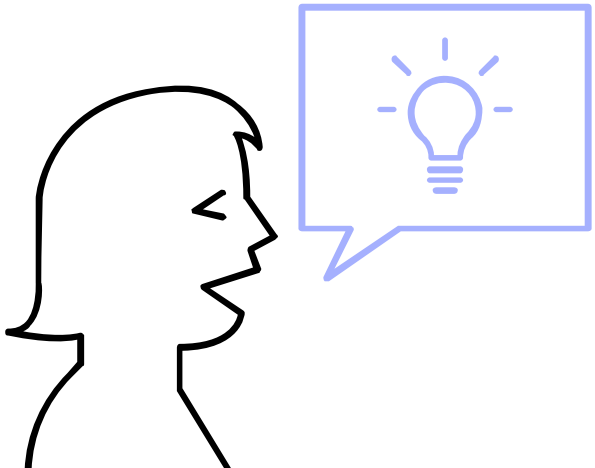




Should models be realistic or aspirational?

Can a growth model be ethical if it ignores impact on users?





- Growth models turn metrics into systems.
- Scenarios reveal risk and potential.
- Retention and virality drive non-linear returns.
- Ethical constraints build resilient companies.



To sum up

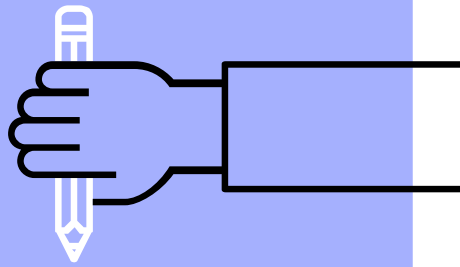
Funnels describe,

loops drive,

metrics measure,

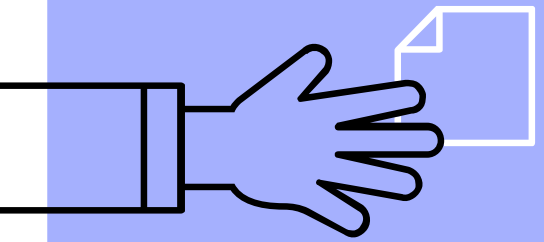
models explain





5.
THE PROJECTS

Now let’s practice !



DATASET 1 : JOLIMOI

ID Order	ID Order Detail	ID Product	ID Product Attribute	Product Quantity	Unit Price Tax Excl	Unit Price Tax Incl	Total Price Tax Excl	Total Price Tax Incl	ID Customer	Total Paid Tax Excl	Created At	Age	Product - ID Product	Product - ID Product →	Customer - ID Custome
78,711	354,084	19992	0	1	9.58	11.5	9.58	11.5	44,498	64.45	June 1, 2021, 12:01 AM	32	SKINCARE	CLEANSING	Île-de-France
78,711	354,087	20048	0	1	12.42	14.9	12.42	14.9	44,498	64.45	June 1, 2021, 12:01 AM	32	SUN PROTECTION	AFTERSUN	Île-de-France
78,711	354,083	19993	0	1	9.58	11.5	9.58	11.5	44,498	64.45	June 1, 2021, 12:01 AM	32	SKINCARE	MOISTURISING	Île-de-France
78,711	354,086	19995	0	1	8.75	10.5	8.75	10.5	44,498	64.45	June 1, 2021, 12:01 AM	32	SKINCARE	MOISTURISING	Île-de-France
78,711	354,082	19994	0	1	10.42	12.5	10.42	12.5	44,498	64.45	June 1, 2021, 12:01 AM	32	SKINCARE	MOISTURISING	Île-de-France
78,711	354,080	19998	0	1	10	12	10	12	44,498	64.45	June 1, 2021, 12:01 AM	32	SKINCARE	MOISTURISING	Île-de-France
78,711	354,085	19991	0	1	6.67	8	6.67	8	44,498	64.45	June 1, 2021, 12:01 AM	32	SKINCARE	CLEANSING	Île-de-France
78,711	354,081	19997	0	1	8.75	10.5	8.75	10.5	44,498	64.45	June 1, 2021, 12:01 AM	32	SKINCARE	PURIFYING	Île-de-France
78,726	354,149	15847	0	1	16.67	20	16.67	20	80,966	119.93	June 1, 2021, 9:38 AM		MAKEUP	NAILS	Île-de-France
78,726	354,148	719	0	1	30	36	30	36	80,966	119.93	June 1, 2021, 9:38 AM		SKINCARE	SERUMS	Île-de-France

Jolimoi is a social selling beauty platform.
Users (called “coaches” or “ambassadors”) recommend products, drive referrals, and generate orders.

DATASET 2 : HEETCH - RIDE DATA

	passenger_id	ride_request_id	country_name	city_id	state	cancel_reason	multiplier	is_painful_cancellation	pre_ride_duration	user_account_created_at	first_ride_request_date	last_ride_request_date	spent_amount_sum_all_time	total_price	estimated_duration	payment_mode	total_estimated_price	estimated_distance	currency	passenger_os_name	stopover_count	coupons_entries_count	passenger_app_version	created_at
0	175af0a-569e-4001-87f5-4774cd9683fb	dcb4d53e-7371-4c21-a180-b2af09e4911b	Alpha	aaa	terminated				400.0	2017-04-22 20:33:25.538	2017-06-30	2019-12-28	26000.00	3600.0	36.0833333333333	cash	3600.00	16.55	c1	Android		6	8.48.0	2017-07-02 02:55:41.211
1	175af0a-569e-4001-87f5-4774cd9683fb	73aea9e4-f0f5-4abc-b60f-a0a650bd272d	Alpha	aaa	canceled_by_driver	passenger_cancel		FALSE		2017-04-22 20:33:25.538	2017-06-30	2019-12-28	26000.00		26.6166666666667	cash	3000.00	13.153	c1	Android		6	8.48.0	2017-07-01 21:49:36.096
2	175af0a-569e-4001-87f5-4774cd9683fb	5143c202-c1ce-4036-a09b-9a4cf9e667ed	Alpha	aaa	canceled_by_driver	passenger_no_show		FALSE		2017-04-22 20:33:25.538	2017-06-30	2019-12-28	26000.00		26.6166666666667	cash	3000.00	13.153	c1	Android		6	8.48.0	2017-07-01 21:51:39.588
3	175af0a-569e-4001-87f5-4774cd9683fb	b852b52e-0214-40ab-a828-23f493afbdc3	Alpha	aaa	terminated				339.0	2017-04-22 20:33:25.538	2017-06-30	2019-12-28	26000.00	4300.0	26.6166666666667	cash	3000.00	13.153	c1	Android		6	8.48.0	2017-07-01 21:58:47.585
4	175af0a-569e-4001-87f5-4774cd9683fb	32806208-dd14-4877-a288-65114c2aa342	Alpha	aaa	terminated				636.0	2017-04-22 20:33:25.538	2017-06-30	2019-12-28	26000.00	1200.0	12.6	cash	1800.00	7.979	c1	Android		6	8.48.0	2017-06-30 22:21:43.135
5	175af0a-569e-4001-87f5-4774cd9683fb	382d219c-5763-4677-bb71-9e7d984bbe7f	Alpha	aaa	canceled_by_driver	cannot_continue		FALSE	4.0	2017-04-22 20:33:25.538	2017-06-30	2019-12-28	26000.00		36.0833333333333	cash	3600.00	16.551	c1	Android		6	8.48.0	2017-07-02 02:35:12.220
6	2c66d21a-c6a1-4680-ac64-68bf8cc04227	dadd9e02-1896-481a-9c72-2fb714a6962c	Alpha	aaa	canceled_by_driver	passenger_cancel		FALSE		2017-01-22 11:38:43.056	2017-01-22	2019-09-01	35200.00		14.3166666666667	cash	2300.00	8.989	c1	Android		2	8.45.1	2017-03-08 12:49:40.911
7	2c66d21a-c6a1-4680-ac64-68bf8cc04227	3387fcdc-41e0-47dd-8ea7-a0f068497827	Alpha	aaa	terminated				663.0	2017-01-22 11:38:43.056	2017-01-22	2019-09-01	35200.00	1600.0	10.9833333333333	cash	1600.00	5.761	c1	Android		2	8.42.0	2017-02-13 20:20:20.978
8	2c66d21a-c6a1-4680-ac64-68bf8cc04227	2c386068-9aad-4f25-8847-b7228bd1c92d	Alpha	aaa	canceled_by_driver	sanitary		FALSE		2017-01-22 11:38:43.056	2017-01-22	2019-09-01	35200.00		10.9833333333333	cash	1600.00	5.758	c1	Android		2	8.42.0	2017-02-03 20:26:03.192
9	2c66d21a-c6a1-4680-ac64-68bf8cc04227	1b6a9415-149f-4898-948f-cd7b7a38f11c	Alpha	aaa	terminated				342.0	2017-01-22 11:38:43.056	2017-01-22	2019-09-01	35200.00	1800.0	11.25	cash	1800.00	6.663	c1	Android		2	8.45.1	2017-03-02 22:02:59.785
10	2c66d21a-c6a1-4680-ac64-68bf8cc04227	b8697873-9ffb-4fbd-9c36-797edb9c94bd	Alpha	aaa	canceled_by_driver	passenger_cancel		FALSE		2017-01-22 11:38:43.056	2017-01-22	2019-09-01	35200.00		5.63333333333333	cash	1000.00	1.37	c1	Android		2	8.42.0	2017-03-02 20:04:29.266
11	2c66d21a-c6a1-4680-ac64-68bf8cc04227	6c441cbc-0fcf-43db-a81e-cb41b511c47e	Alpha	aaa	canceled_by_driver	cannot_pickup_passenger		FALSE		2017-01-22 11:38:43.056	2017-01-22	2019-09-01	35200.00		5.63333333333333	cash	1000.00	1.37	c1	Android		2	8.42.0	2017-03-02 19:59:38.257
12	2c66d21a-c6a1-4680-ac64-68bf8cc04227	3058d97e-20dd-486d-b24a-971cafa1798c	Alpha	aaa	terminated				346.0	2017-01-22 11:38:43.056	2017-01-22	2019-09-01	35200.00	1600.0	11.2333333333333	cash	1600.00	5.76	c1	Android		2	8.45.1	2017-03-09 13:45:55.710
13	2c66d21a-c6a1-4680-ac64-68bf8cc04227	0916caf1-688f-4edd-8988-ba2f511acfc8	Alpha	aaa	canceled_by_driver	passenger_no_answer		FALSE		2017-01-22 11:38:43.056	2017-01-22	2019-09-01	35200.00		5.63333333333333	cash	1000.00	1.37	c1	Android		2	8.42.0	2017-03-02 19:56:16.361
14	2c66d21a-c6a1-4680-ac64-68bf8cc04227	898f8586-392c-4429-a7ed-3b06bc1246b9	Alpha	aaa	canceled_by_user	cannot_continue_pagr		FALSE		2017-01-22 11:38:43.056	2017-01-22	2019-09-01	35200.00		10.7333333333333	cash	1300.00	4.102	c1	Android		2	8.45.1	2017-03-08 20:16:24.621

Heetch is a ride hailing application. It's a marketplace with a demand side (passenger) and a supply side (driver)

DATASET 3 : HEETCH - AD SPEND

	ch_01	ch_02	ch_03	ch_04	ch_05	ch_06	ch_07	ch_08	ch_09	ch_10	ch_11	ch_12	ch_13	ch_14	ch_15	ch_16	ch_17	ch_18	rides_requested	new_users	weekly_redeemed_amount	t
0	5978.712	1324.5428	0.0	0.0	2312.6643	100.8	998.63	0.0	1088.132	0.0	0.0	0.0	0.0	0.0	0.0	0.0	43.2	3561.13420348149	126601	5459	1106163	1
1	18663.695	3882.1863	0.0	0.0	11010.4575	214.4	3034.45	0.0	3670.724	0.0	0.0	0.0	0.0	0.0	0.0	0.0	123.2	5547.18544810882	258050	9647	5421460	2
2	9772.1135	3735.9187	0.0	494.66	13642.8175	340.8	2538.77	1012.18	3299.0041	0.0	0.0	0.0	0.0	0.0	0.0	0.0	156.8	3593.77281582999	256980	9435	4057831	3
3	7905.34150000001	3369.287	0.0	1259.18	10096.8081	361.6	2418.06	297.53	2875.9567	0.0	0.0	0.0	0.0	0.0	0.0	0.0	16.0	3872.81216155587	237608	8634	2907957	4
4	7304.1445	3127.4005	0.0	1258.69	12723.5325	84.8	1129.89	11.73	2834.4051	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	5179.80756369285	243950	8754	3010726	5
5	10443.38	7974.7229	6541.5	1262.98	10968.3331	120.0	2053.08	144.05	2874.7435	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	5083.55846796682	282855	10010	2804727	6
6	10950.185	7756.3115	14566.0	1259.83	11344.9408	267.2	2052.78	17.27	3050.967	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3725.4644950316	300255	10876	4387815	7
7	13880.914	9074.2687	14929.0	1258.57	14096.7037	163.2	2629.11	1.27	3259.3511	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4850.9518351706	349260	13517	4688313	8
8	12256.7805	8525.4903	10583.0	1262.22	9310.7964	374.4	2880.34	0.0	5456.4854	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4791.89153634457	279741	10756	7810112	9
9	12212.5515	8401.4597	9712.0	1260.16	12191.2214	256.0	2910.18	0.0	4318.4448	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6087.43430476021	322331	11267	7618732	10
10	13331.03	8525.5513	12299.0	1094.62	12638.5708	326.4	2724.59	0.0	4037.8921	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4097.09990282476	325694	11043	5465429	11
11	15151.664	8399.0937	12240.5	1127.58	11567.1872	1140.8	3675.01	0.0	4045.0308	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3907.76408507215	308074	10663	3594754	12
12	9767.3295	8218.283	9940.0	967.07	11199.9535	617.6	1912.34	0.0	3844.0271	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3952.32994858604	287825	10423	4081331	13

Heetch is a ride hailing application. It's a marketplace with a demand side (passenger) and a supply side (driver)